



VMWARE VSPHERE 7.0: INSTALL, CONFIGURE, MANAGE



Lab 03: Working with vCenter Server

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Introduction

In this lab, you will install and use vCenter Server Appliance.

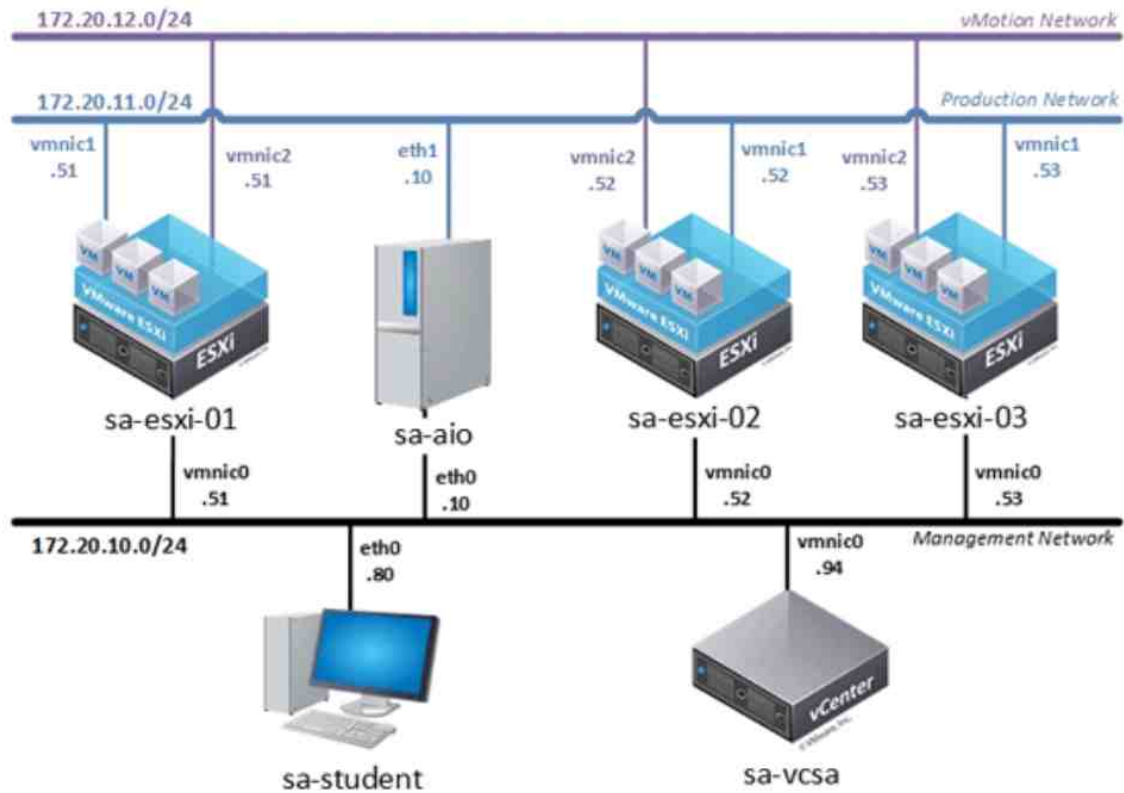


This lab is supplemental and takes approximately two hours to complete. Be prepared to wait up to 20 minutes in this lab for task 2. It is also advised that you make the reservation for 3 hours.

Objectives

-) Deploy vCenter Server Appliance
-) Complete the vCenter Server Appliance deployment
-) Access the vCenter Server Appliance
-) License the vCenter Server Appliance

Lab Topology



Lab Settings

The information in the table below will be needed in order to complete the lab. The task sections below provide details on the use of this information.

Virtual Machine	IP Address	Account	Password
sa-esxi-01	eth0: 172.20.10.51	root	Train1ng\$
sa-esxi-02	eth0: 172.20.10.52	root	Train1ng\$
sa-esxi-03	eth0: 172.20.10.53	root	Train1ng\$
sa-aio	eth0: 172.20.10.10	sysadmin	vmware123
sa-student	eth0: 172.20.10.80	sysadmin	vmware123
sa-vcsa	eth0: 172.20.10.94	sysadmin@vclass.local	vmware123

1 Prepare the Lab Environment for vCenter Server Install

In this task, you will prepare the lab environment to start the process of installing vCSA.



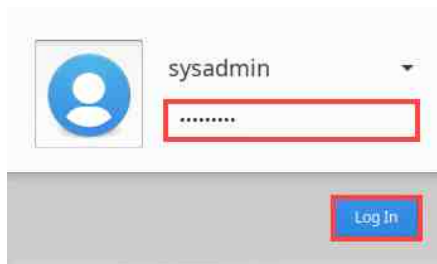
Please allow sufficient time to complete this lab. You will be completing a full install of a vCenter Server Appliance which will take approximately 2 hours from beginning to finish. There will be two tasks that will take 20 minutes to complete.

1. Launch the sa-student virtual machine to access the graphical login screen.

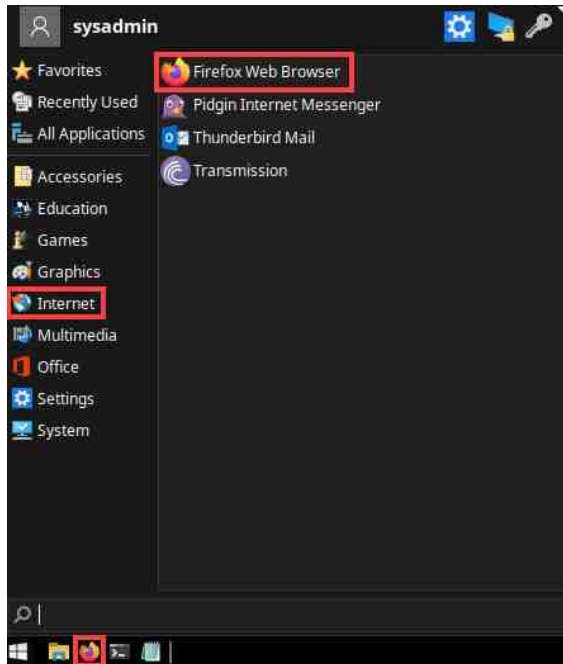


To launch the console window for a virtual machine, you may access by either clicking on the machine's graphic image from the topology page or by clicking on the machine's respective tab from the navigation bar.

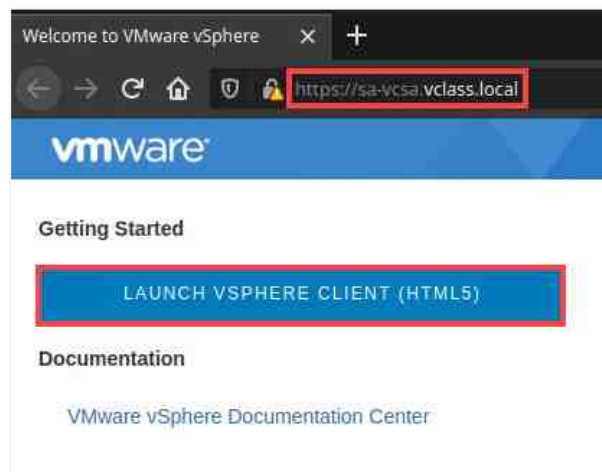
2. Log in as sysadmin using the password vmware123.



3. Launch the Mozilla Firefox web browser by either clicking on the icon shortcut found on the bottom toolbar or by navigating to Start Menu > Internet > Firefox Web Browser.

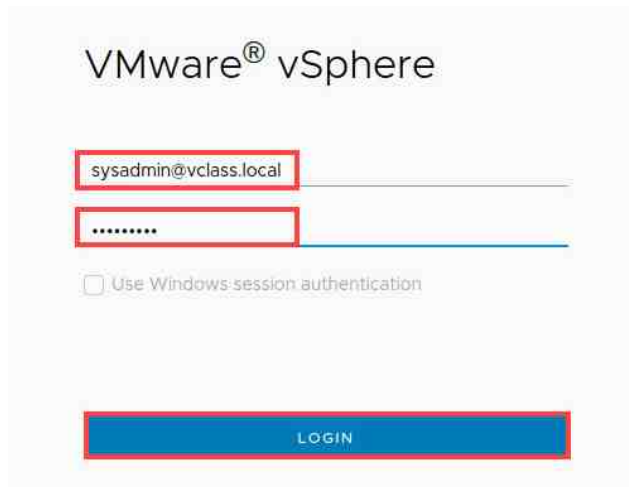


4. Notice the homepage is automatically configured to load the URL address of the VMware vCenter Server Appliance (sa-vcsa virtual machine). Click on the LAUNCH VSPHERE CLIENT (HTML5) button to launch the vSphere Client.



If you cannot successfully load the VMware Getting Started webpage, then please wait an additional 3-5 minutes and refresh the page to continue. During this time, the vCenter Server Appliance is still booting up and requires extra time to initialize.

5. In the Username text field, type `sysadmin@vclass.local` and in the Password text field, type `vmware123`. Click on LOGIN.

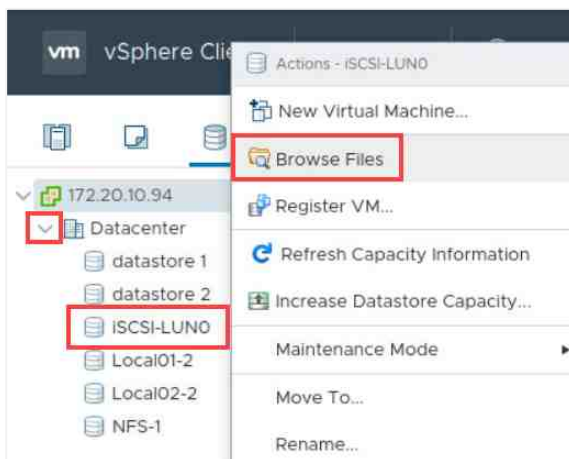
**Please Note**

You may ignore the “browser-OS combination” warning message presented on the VMware vCenter Single Sign-On page and continue moving forward with the lab.

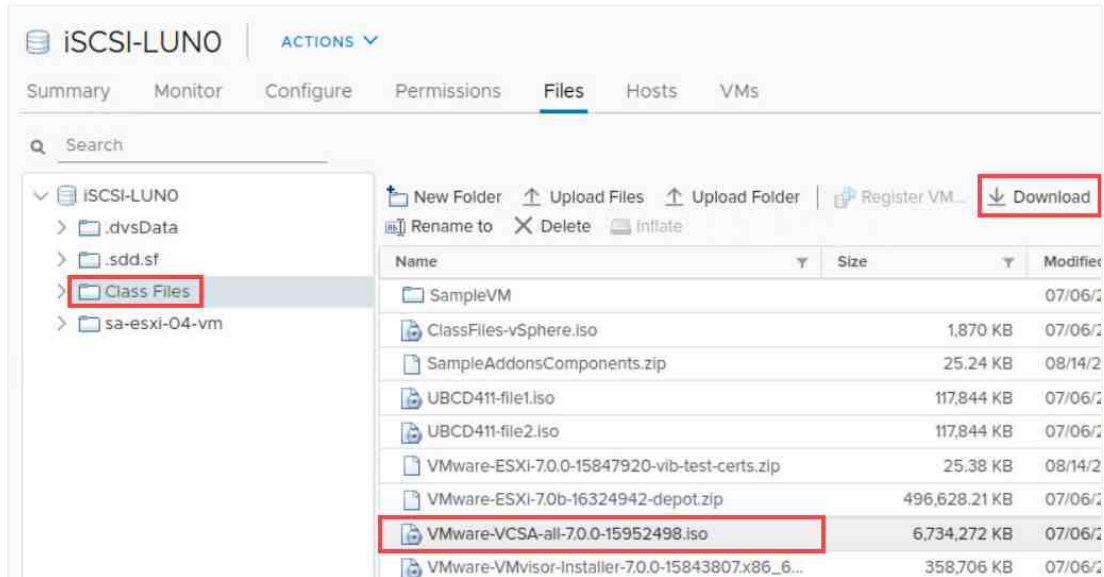
6. In the navigator pane, click on the Storage tab.



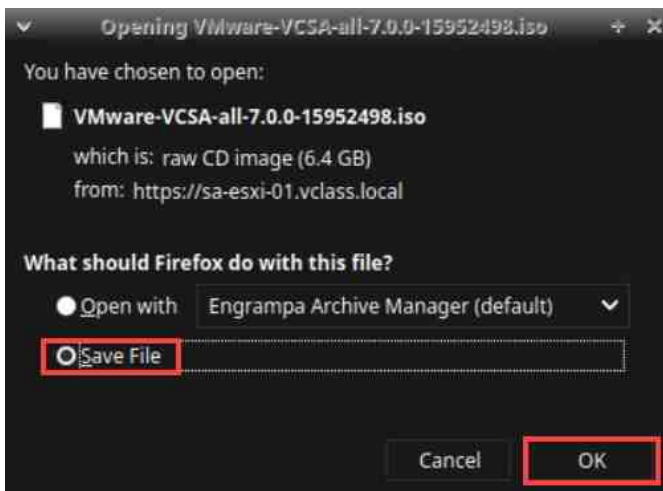
7. Expand the Datacenter object in the navigator pane, right-click on iSCSI-LUN0 and select Browse Files.



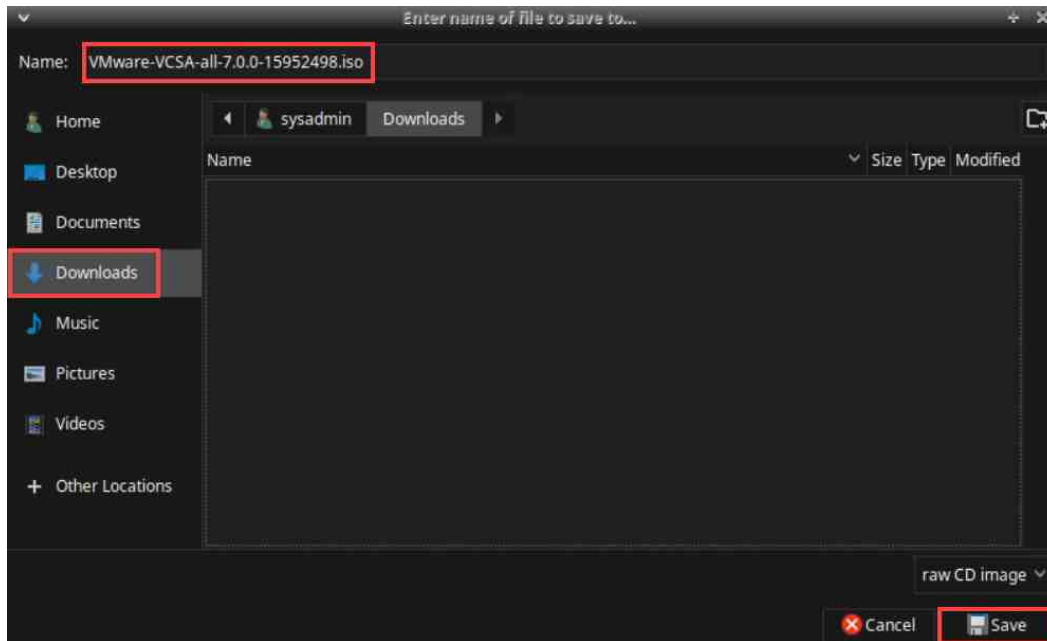
8. In the main workspace, access the Class Files folder, select the VMware-VCSA-all-7.0.0-15952498.iso file, and then click on Download.



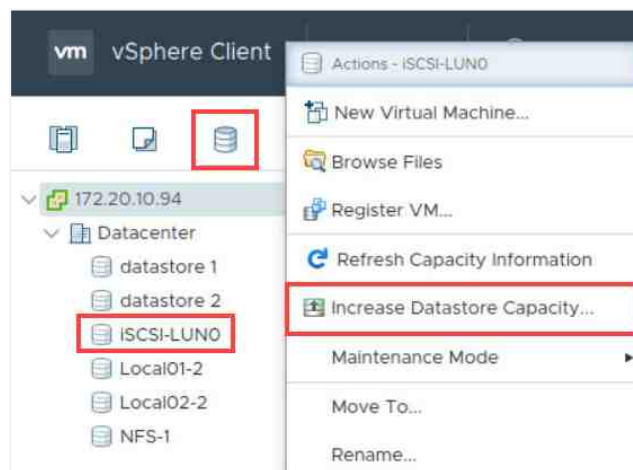
9. In the download dialog box, select Save File and click OK.



10. In the Enter name of file to save to window, verify VMware-VCSA-all-7.0.0-15952498.iso is showing in the Name Field. Select Downloads and click Save.



11. Before you start the vCenter Server Appliance deployment, you need to increase disk space for the minimum required storage space of 25GB. Navigate back to the vSphere Client and select the Storage tab. Right-click on iSCSI-LUN0 and select Increase Datastore Capacity.



12. In the Increase Datastore Capacity window, on the Select Device step, select the IET iSCSI Disk LUN 0 option and click Next.

Increase Datastore Capacity

1 Select Device
2 Specify Configuration
3 Ready To Complete

Select Device
Select disk/LUN to be used to increase size of the datastore.

Name	LUN	Capacity	Hardware...	Drive T...	Se
IET iSCSI Disk (t10.94544...	0	23.50 GB	Unknown	HDD	5'

1 items

CANCEL **BACK** **NEXT**

13. On the Specify Configuration step, choose Use "Free space 15.5 GB" to expand the datastore in the Partition Configuration dropdown menu and click Next.

Increase Datastore Capacity

✓ **1 Select Device**
2 Specify Configuration
3 Ready To Complete

Specify Configuration
Review the disk layout and specify partition configuration.

Partition Configuration: Use "Free space 15.5 GB" to expand the datastore

Increase Size by: 15.5 GB

ISCSI-LUN0 (VMFS): 8.0 GB | Empty: 15.5 GB

ISCSI-LUN0 (VMFS)

Info The datastore already occupies one or more extents on this device. Selecting free space adjacent to an extent in the datastore will expand that extent. Selecting any other option will add a new extent to the datastore.

CANCEL **BACK** **NEXT**

14. On the Ready to Complete step, review the information and click Finish.

Increase Datastore Capacity

✓ 1 Select Device

✓ 2 Specify Configuration

3 Ready To Complete

Ready To Complete

Review your settings for increasing the size of this datastore.

General:

Name

ISCSI-LUN0

Increase Datastore Size by

15.50 GB

Future Datastore Size

50.50 GB

Device and Formatting:

Disk/LUN

IET iSCSI Disk
(t10.945445000000000000F64696E603000000000000
0000000000)

Partition Format

GPT

VMFS Version

VMFS 5.81

Maximum File Size

62.94 TB

Block Size

1 MB

CANCEL

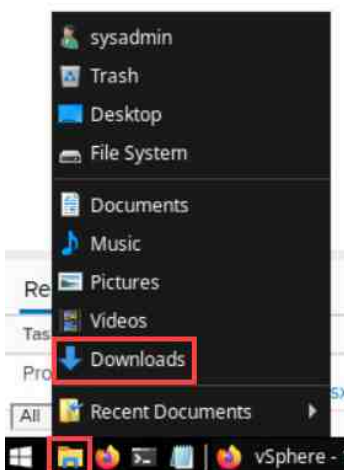
BACK

FINISH

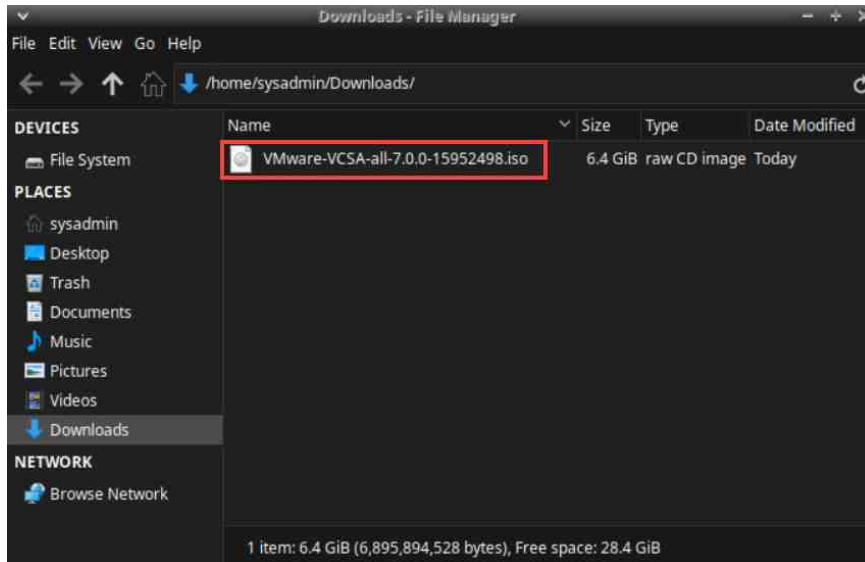
15. Monitor the Recent Tasks pane and verify the Datastore is expanded. Once complete, continue to the next step.

Recent Tasks		Alarms							
Task Name	Target	Status	Details	Initiator	Target	Queued For	Start Time	Completion Time	Server
Process VMFS datastore updates	 sa-esxi-02.vcl...	 Completed		System		5 ms	09/28/2020, 11:22:06 PM	09/28/2020, 11:22:07 PM	172.20.10.94
Expand VMFS datastore	 sa-esxi-01.vcl...	 Completed		VCLASS.LOCAL\sy...		2 ms	09/28/2020, 11:22:01 PM	09/28/2020, 11:22:06 PM	172.20.10.94

16. Once the datastore is expanded, access the downloaded file by first clicking on the Places icon in the bottom toolbar. Click Downloads.

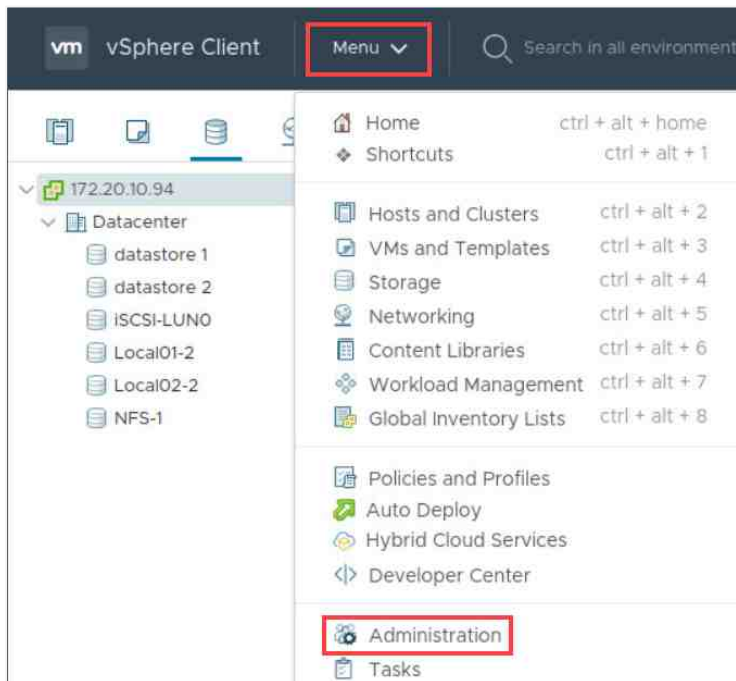


17. Verify the VMware-VCSA-all-7.0.0-15952498.iso is showing in the Downloads – File Manager.

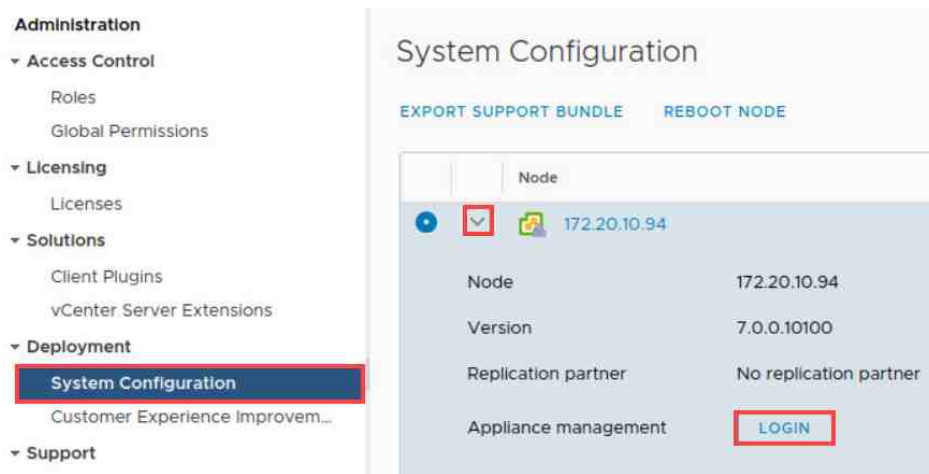


18. Minimize the Downloads – File Manager window.

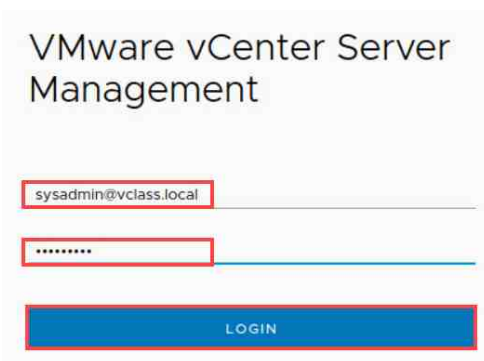
19. In the vSphere Client, select Menu and click Administration.



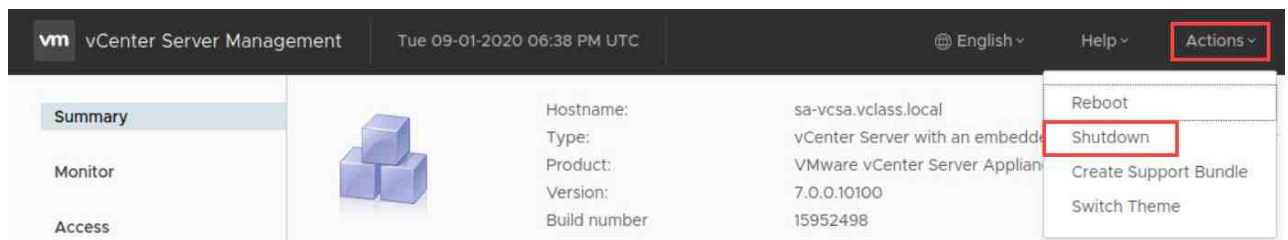
20. In the left pane, select System Configuration. In the System Configuration window, expand 172.20.10.94 and click LOGIN.



21. On the VMware vCenter Server Management window, enter `sysadmin@vclass.local` as the user and `vmware123` for the password. Click LOGIN.



22. In the vCenter Server Management window, expand Actions and click Shutdown.



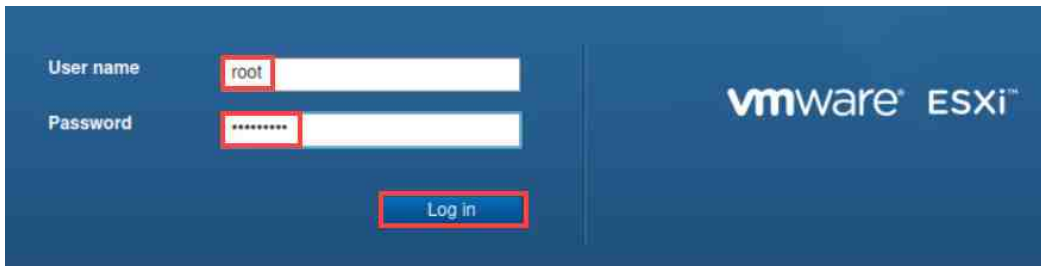
23. On the System Shutdown window, click YES.



24. In this step, you will add iSCSI-LUN0 to sa-esxi-03. In the Firefox window, type `https://172.20.10.53` in the address bar and press Enter.



25. On the VMware ESXi host login screen, type `root` for the user and `Training$` for the Password. Click Log in.



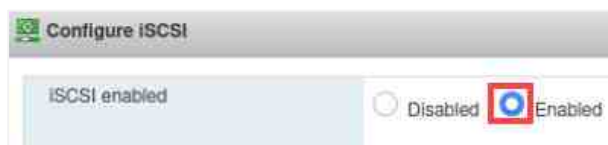
26. In the Navigator pane, select Storage.



27. On the sa-esxi-03.netlab.local – Storage window, select the Adapters tab. Click on Software iSCSI.



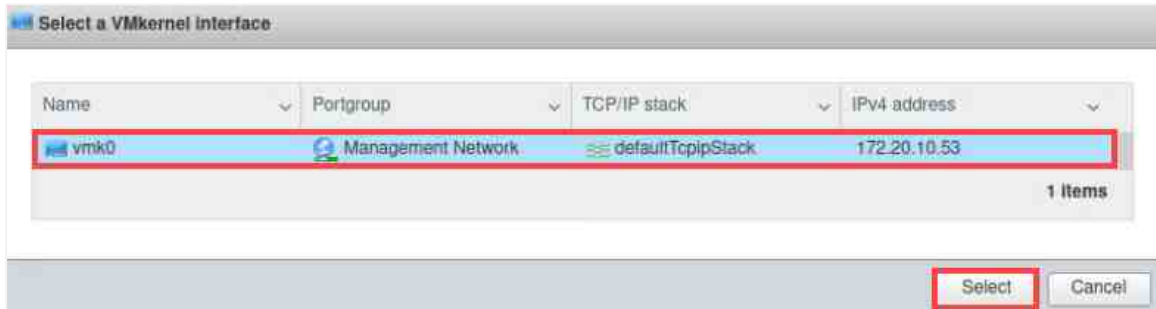
28. On the Configure iSCSI window, click Enabled to configure iSCSI.



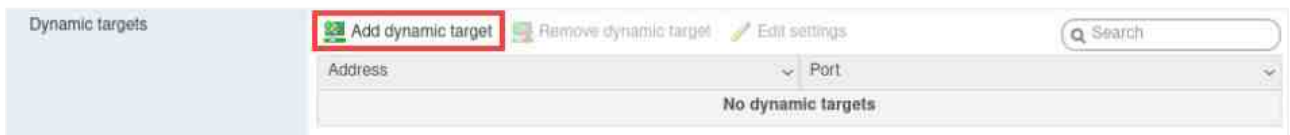
29. On the Network Port bindings option, click Add port binding.



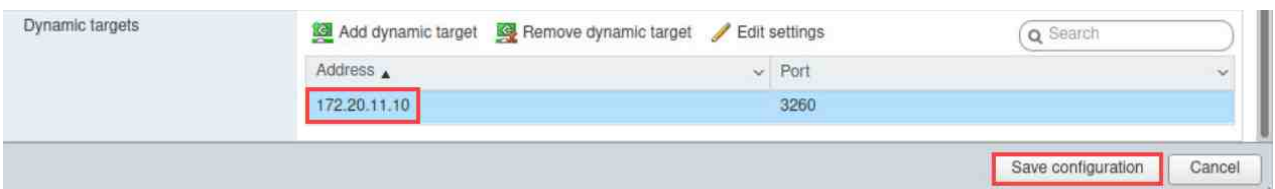
30. On the Select a VMkernel interface window, select vmk0. Click Select.



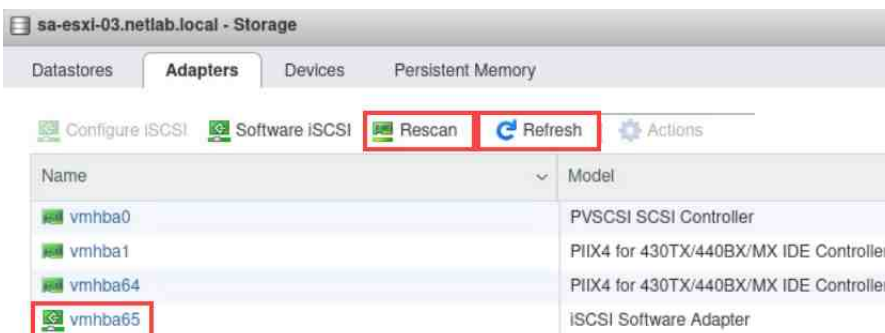
31. On the Dynamic targets option, click Add dynamic target.



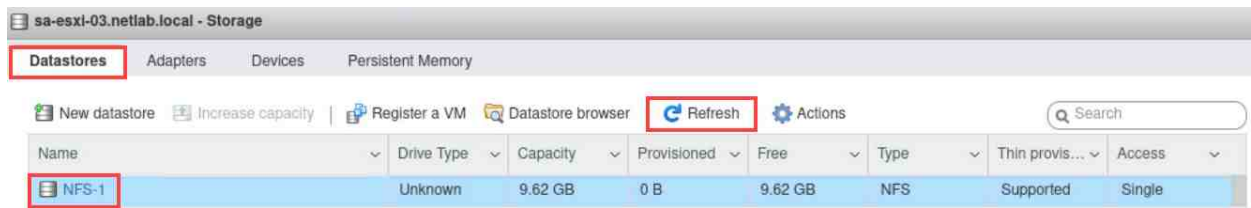
32. In the Dynamic targets Address field, type 172.20.11.10 and click Save configuration.



33. On the Adapters tab, click Rescan. If the vmhba65 iSCSI adapter is not showing after rescanning, click Refresh. Verify vmhba65 appears.



34. Click on the Datastores tab, select NFS-1 and click Refresh.



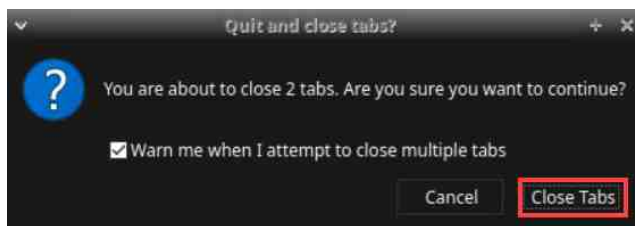
35. Verify iSCSI-LUN0 appears.



36. Close Mozilla Firefox by clicking the X in the upper-right corner.



37. In the Quit and close tabs window, click Close Tabs.



38. Continue to the next task.

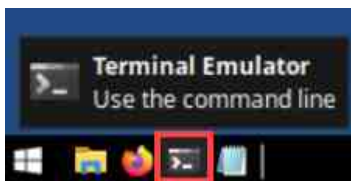
2 Deploy vCenter Server Appliance

In this task, you will go through the process of deploying a VMware vCenter Server Appliance on your sa-esxi-03 host.



There are two stages to deploy vCenter Server. Stage 1 will take approximately 20 minutes and Stage 2 will also take approximately 20 minutes to complete. This task will take 40 minutes to complete. Be prepared to wait patiently until the full install has completed.

1. On the sa-student machine, launch a terminal by clicking on the Terminal Emulator icon on the bottom toolbar.



2. Change to the superuser by entering the command below, followed by pressing the Enter key. When prompted for a password, type `vmware123` and press Enter.

```
sysadmin@sa-student-01:~$ sudo su
```

```
sysadmin@sa-student-01:~$ sudo su
[sudo] password for sysadmin:
root@sa-student-01:/home/sysadmin#
```

3. Make a directory called `vcasa` to mount the iso. `/mnt/vcasa`

```
root@sa-student-01:/home/sysadmin# mkdir /mnt/vcasa
```

```
root@sa-student-01:/home/sysadmin# mkdir /mnt/vcasa
root@sa-student-01:/home/sysadmin#
```

4. Mount the vCSA installer ISO to the `/mnt/vcasa/` path.

```
root@sa-student-01:/home/sysadmin# mount /home/sysadmin/Downloads/VMware-
VCSA-all-7.0.0-15952498.iso /mnt/vcasa/
```

```
root@sa-student-01:/home/sysadmin# mount /home/sysadmin/Downloads/VMware-VCSA-all-7.0.0-15952498.iso /mnt/vcasa
mount: /mnt/vcasa: WARNING: device write-protected, mounted read-only.
root@sa-student-01:/home/sysadmin#
```

5. Verify that the ISO contents have been successfully mounted to the path.

```
root@sa-student-01:/home/sysadmin# ls /mnt/vcsa
```

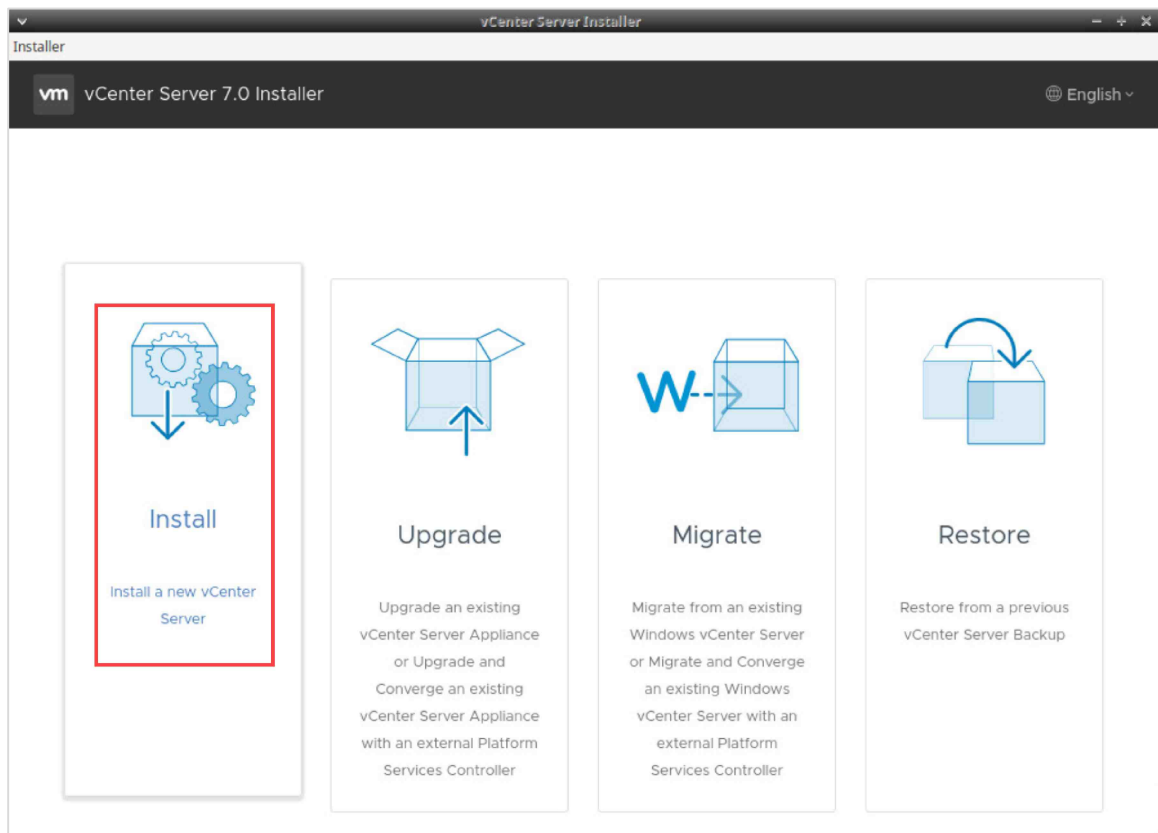
```
root@sa-student-01:/home/sysadmin# ls /mnt/vcsa/
dbschema          readme-de.txt      readme-fr.txt      readme-ko.txt      readme-zh-CN.txt  umds  vcsa-cli-installer
migration-assistant readme-es.txt      readme-ja.txt      readme.txt         readme-zh-TW.txt  vcsa  vcsa-ui-installer
root@sa-student-01:/home/sysadmin#
```

6. Initiate the VMware vCSA installer application.

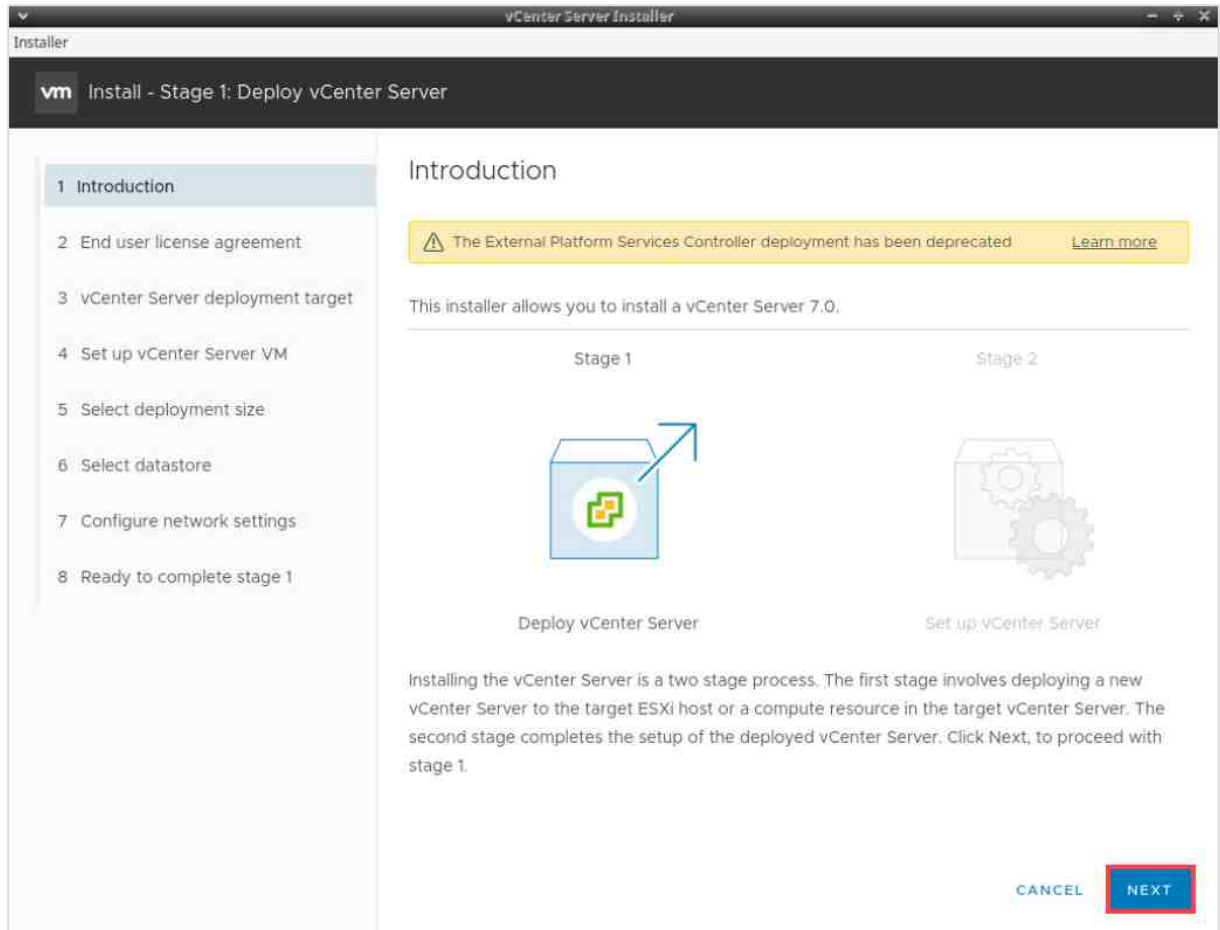
```
root@sa-student-01:/home/sysadmin# /mnt/vcsa/vcsa-ui-
installer/lin64/installer
```

```
root@sa-student-01:/home/sysadmin# /mnt/vcsa/vcsa-ui-installer/lin64/installer
```

7. Notice the vCenter Server Appliance Installer window appears. Click Install.



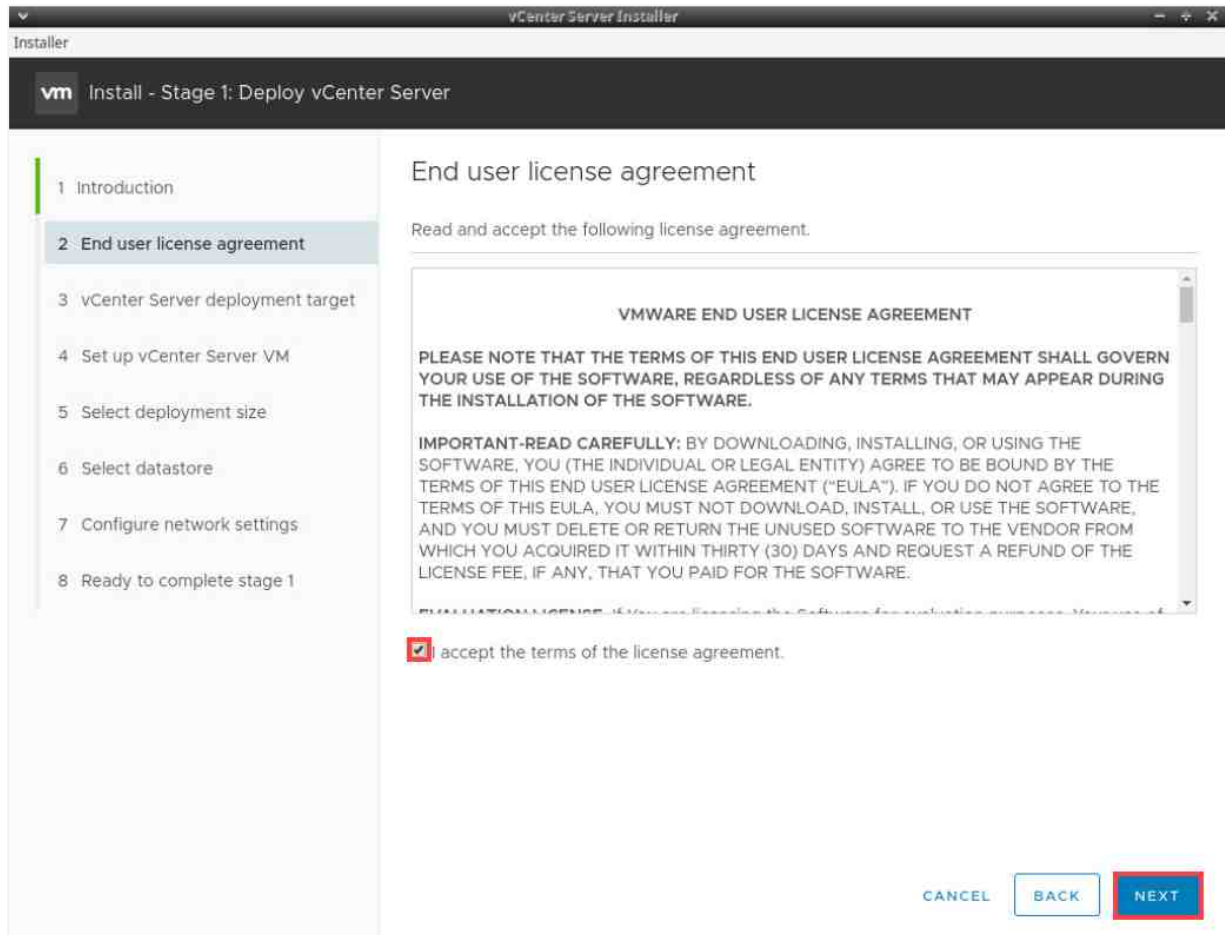
8. On the Introduction page, click Next to continue.



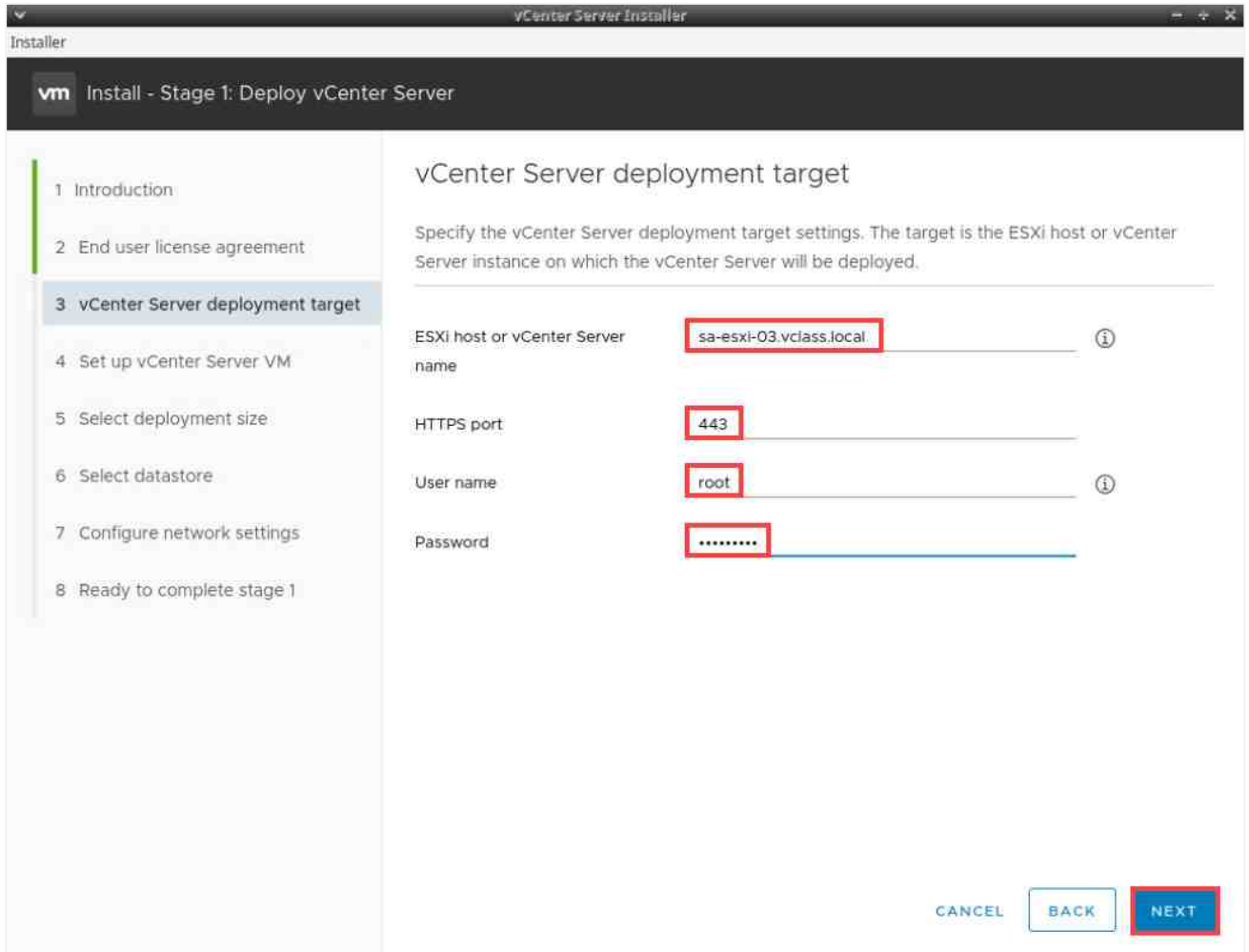
**Please
Note**

Notice the informational message that the External Platform Services Controller has been deprecated. Therefore, it has carried over to v7 and will not be available in future releases. You may ignore the message.

9. On the End user license agreement page, select the I accept the terms of the license agreement checkbox and click Next.



10. On the vCenter Server deployment target page, enter the following information into their appropriate text fields:
- ESXi host or vCenter Server name: `sa-esxi-03.vclass.local`
 - HTTPS port: 443
 - Username: `root`
 - Password: `Training$`
 - Click Next.



vCenter Server Installer

Installer

vm Install - Stage 1: Deploy vCenter Server

1 Introduction

2 End user license agreement

3 vCenter Server deployment target

4 Set up vCenter Server VM

5 Select deployment size

6 Select datastore

7 Configure network settings

8 Ready to complete stage 1

vCenter Server deployment target

Specify the vCenter Server deployment target settings. The target is the ESXi host or vCenter Server instance on which the vCenter Server will be deployed.

ESXi host or vCenter Server name

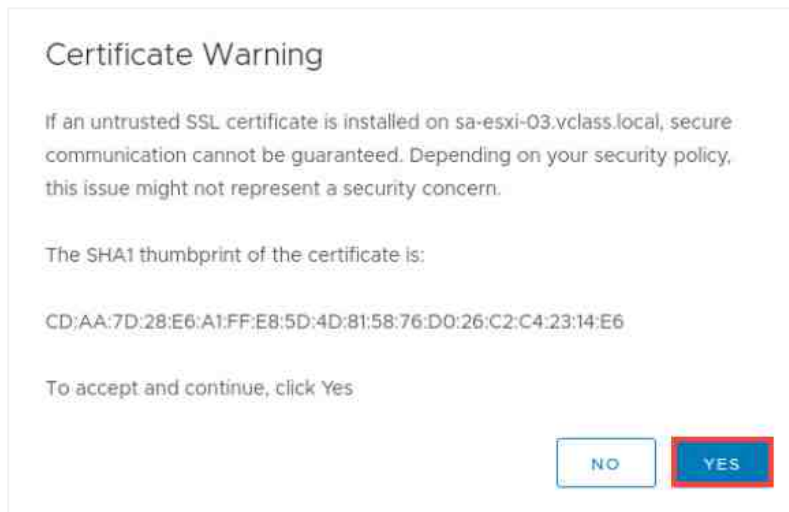
HTTPS port

User name

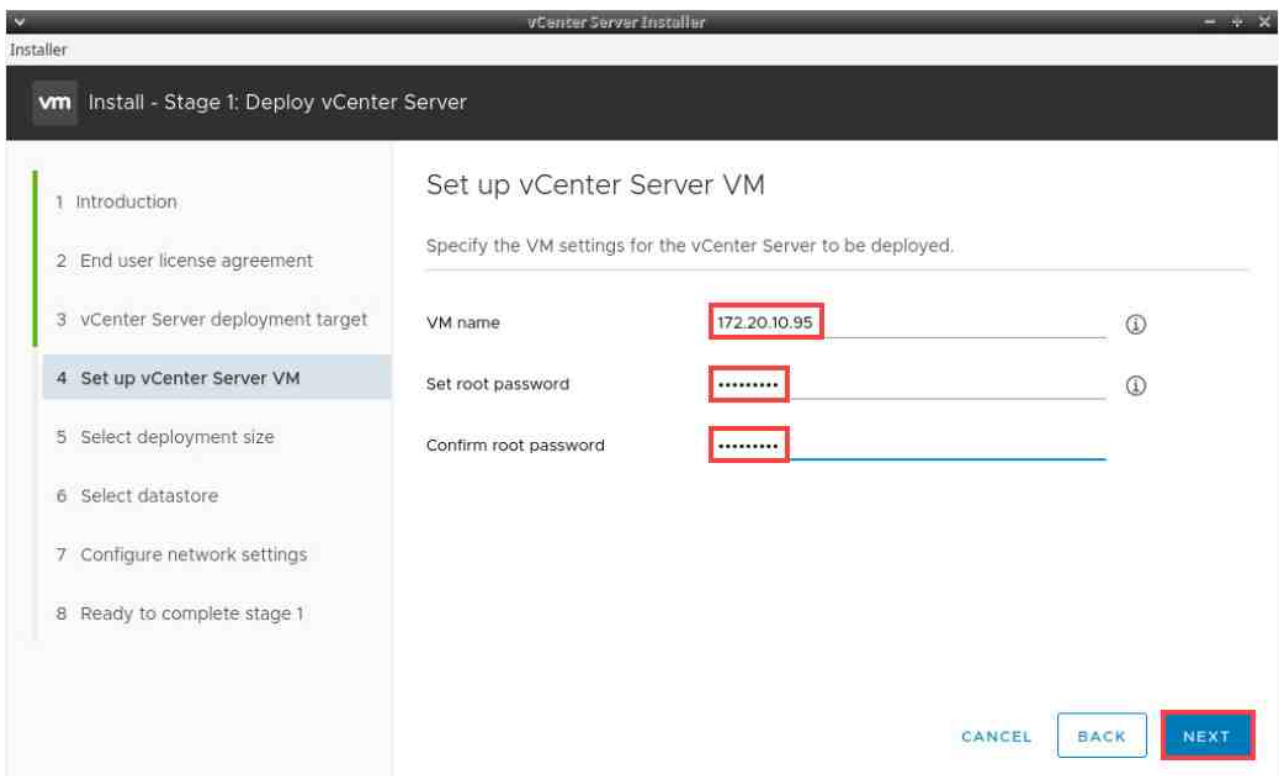
Password

CANCEL BACK NEXT

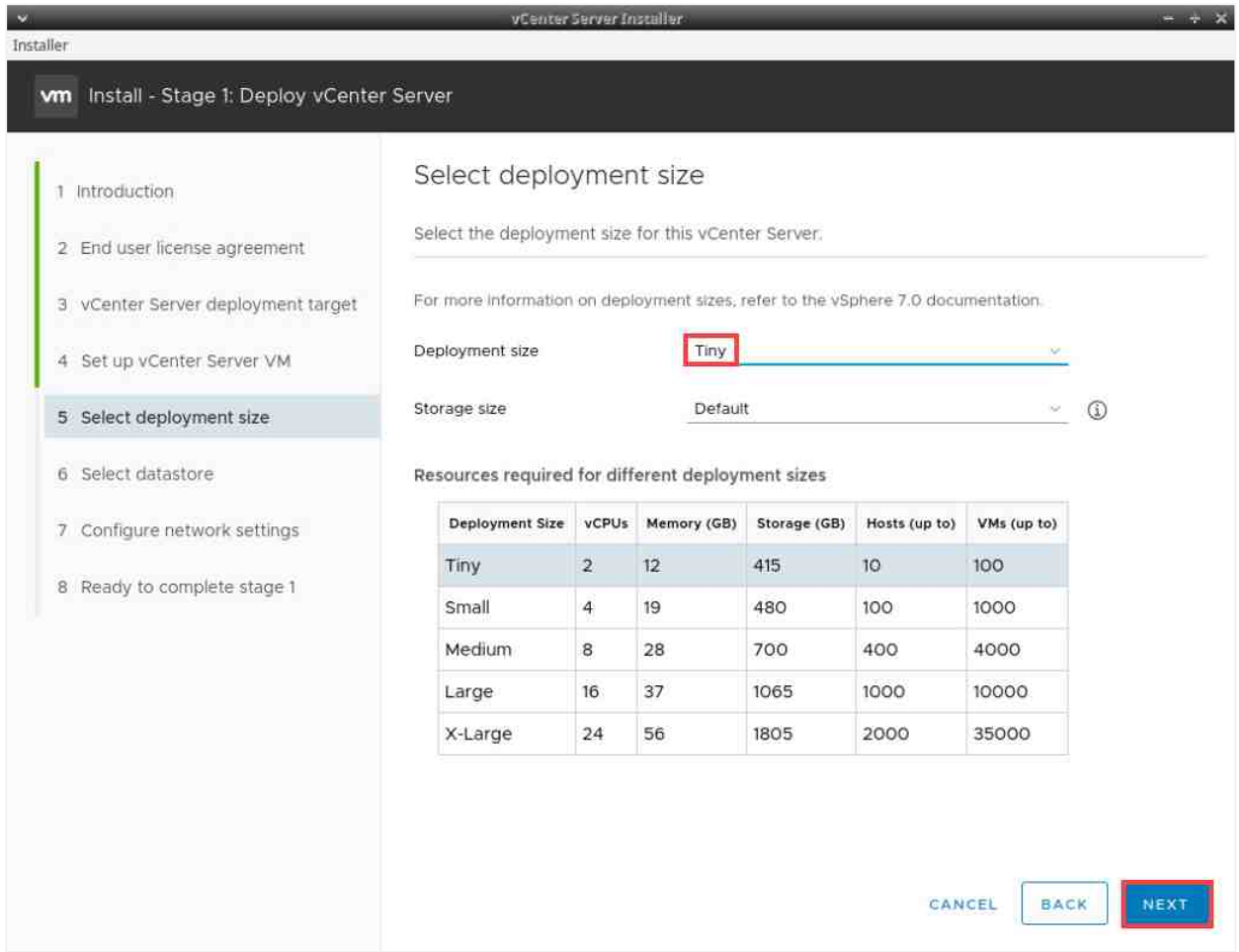
11. Notice the Certificate Warning dialog box, review and click Yes to continue.



12. On the Set up vCenter Server VM page, enter 172.20.10.95 in the VM name field and enter Train1ng\$ for both the Root password and the Confirm root password text fields. Click Next.



13. On the Select deployment size page, accept the default, Tiny, and click Next.



The screenshot shows the vCenter Server Installer window at Stage 1: Deploy vCenter Server. The left sidebar contains a list of steps, with '5 Select deployment size' highlighted. The main area is titled 'Select deployment size' and includes instructions to select a deployment size for the vCenter Server. It provides a link to vSphere 7.0 documentation for more information on deployment sizes. Below this, there are two dropdown menus: 'Deployment size' set to 'Tiny' and 'Storage size' set to 'Default'. A table titled 'Resources required for different deployment sizes' is displayed, showing requirements for Tiny, Small, Medium, Large, and X-Large sizes. At the bottom right, there are three buttons: 'CANCEL', 'BACK', and 'NEXT'.

Installer

vm Install - Stage 1: Deploy vCenter Server

1 Introduction

2 End user license agreement

3 vCenter Server deployment target

4 Set up vCenter Server VM

5 Select deployment size

6 Select datastore

7 Configure network settings

8 Ready to complete stage 1

Select deployment size

Select the deployment size for this vCenter Server.

For more information on deployment sizes, refer to the vSphere 7.0 documentation.

Deployment size **Tiny**

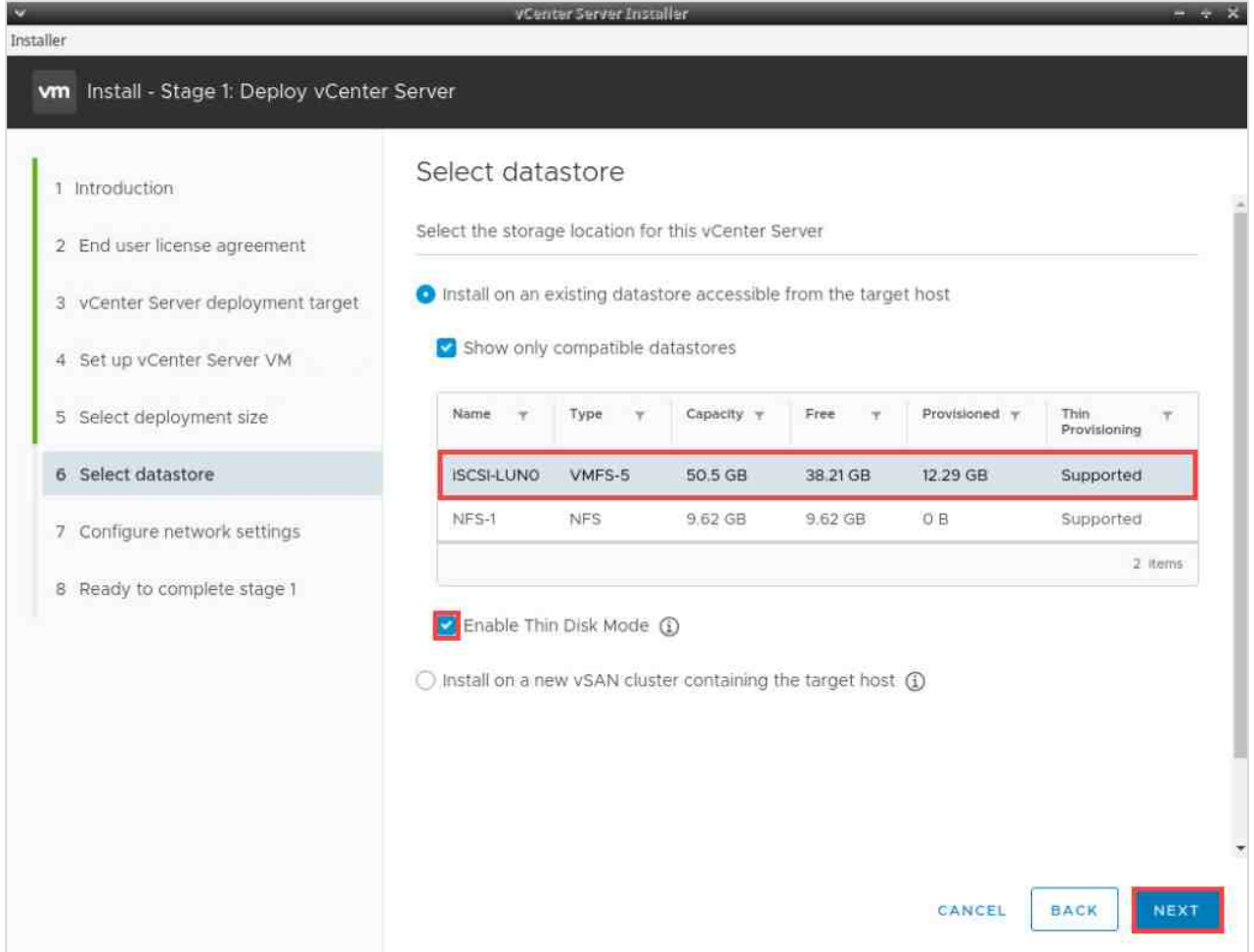
Storage size Default ⓘ

Resources required for different deployment sizes

Deployment Size	vCPUs	Memory (GB)	Storage (GB)	Hosts (up to)	VMs (up to)
Tiny	2	12	415	10	100
Small	4	19	480	100	1000
Medium	8	28	700	400	4000
Large	16	37	1065	1000	10000
X-Large	24	56	1805	2000	35000

CANCEL BACK **NEXT**

14. On the Select datastore page, select iSCSI-LUN0 as the deployment datastore and check Enable Thin Disk Mode. Click Next.



The screenshot shows the 'vCenter Server Installer' window at the 'Select datastore' stage. The left sidebar lists the installation steps, with '6. Select datastore' highlighted. The main area shows options to install on an existing datastore or a new vSAN cluster. The 'Show only compatible datastores' checkbox is checked, displaying a table of available datastores. The 'iSCSI-LUN0' datastore is highlighted with a red box. Below the table, the 'Enable Thin Disk Mode' checkbox is also checked and highlighted with a red box. The 'NEXT' button at the bottom right is highlighted with a red box.

Installer

vm Install - Stage 1: Deploy vCenter Server

1 Introduction

2 End user license agreement

3 vCenter Server deployment target

4 Set up vCenter Server VM

5 Select deployment size

6 Select datastore

7 Configure network settings

8 Ready to complete stage 1

Select datastore

Select the storage location for this vCenter Server

☒ Install on an existing datastore accessible from the target host

☒ Show only compatible datastores

Name	Type	Capacity	Free	Provisioned	Thin Provisioning
iSCSI-LUN0	VMFS-5	50.5 GB	38.21 GB	12.29 GB	Supported
NFS-1	NFS	9.62 GB	9.62 GB	0 B	Supported

2 items

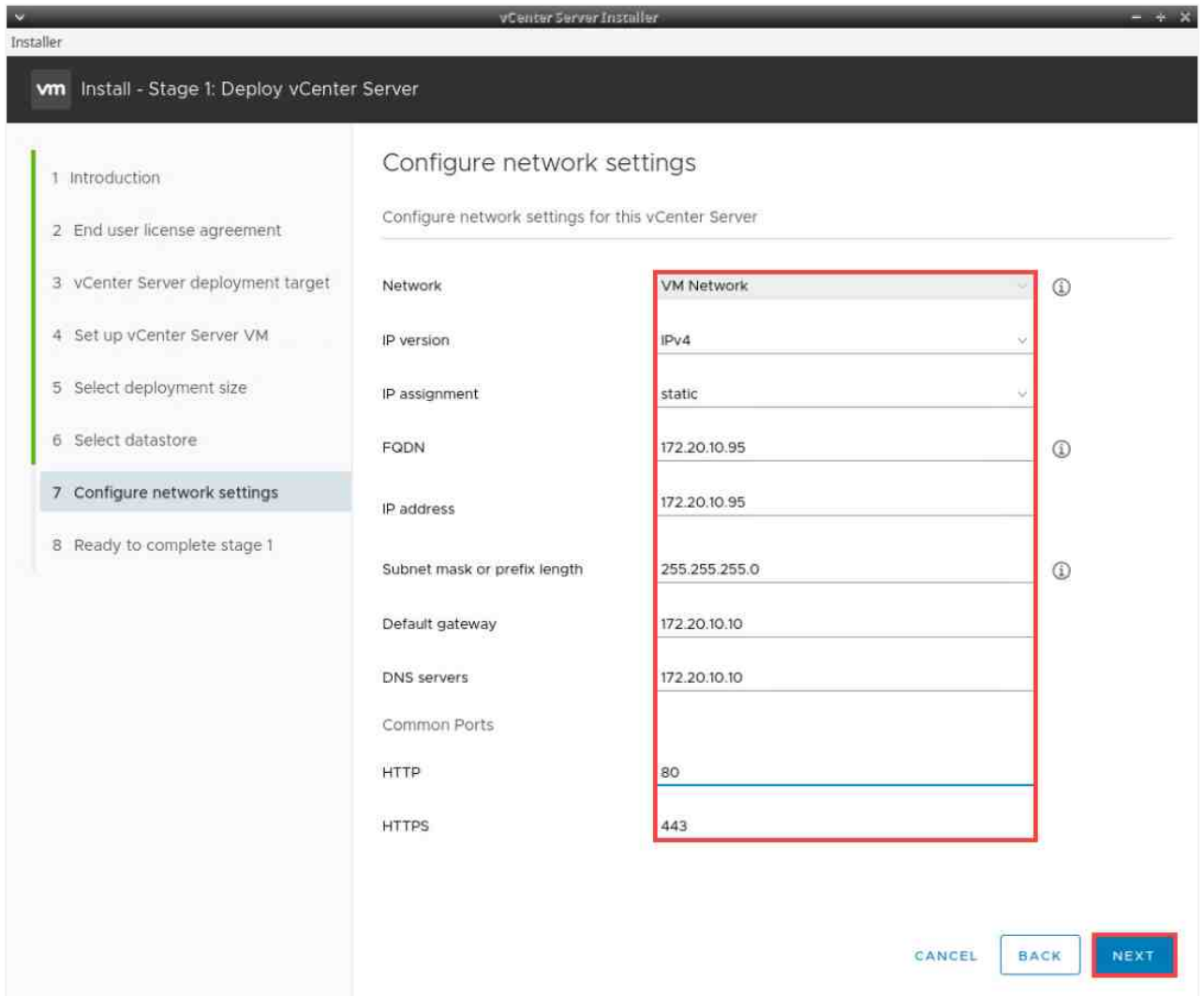
☒ Enable Thin Disk Mode ⓘ

☐ Install on a new vSAN cluster containing the target host ⓘ

CANCEL BACK **NEXT**

15. On the Configure network settings page, configure the following:

- a. Network: VM Network
- b. IP version: IPv4
- c. IP assignment: static
- d. FQDN: 172.20.10.95
- e. IP address: 172.20.10.95
- f. Subnet mask or prefix: 255.255.255.0
- g. Default gateway: 172.20.10.10
- h. DNS servers: 172.20.10.10
- i. Click Next.



vCenter Server Installer

Installer

vm Install - Stage 1: Deploy vCenter Server

1 Introduction

2 End user license agreement

3 vCenter Server deployment target

4 Set up vCenter Server VM

5 Select deployment size

6 Select datastore

7 Configure network settings

8 Ready to complete stage 1

Configure network settings

Configure network settings for this vCenter Server

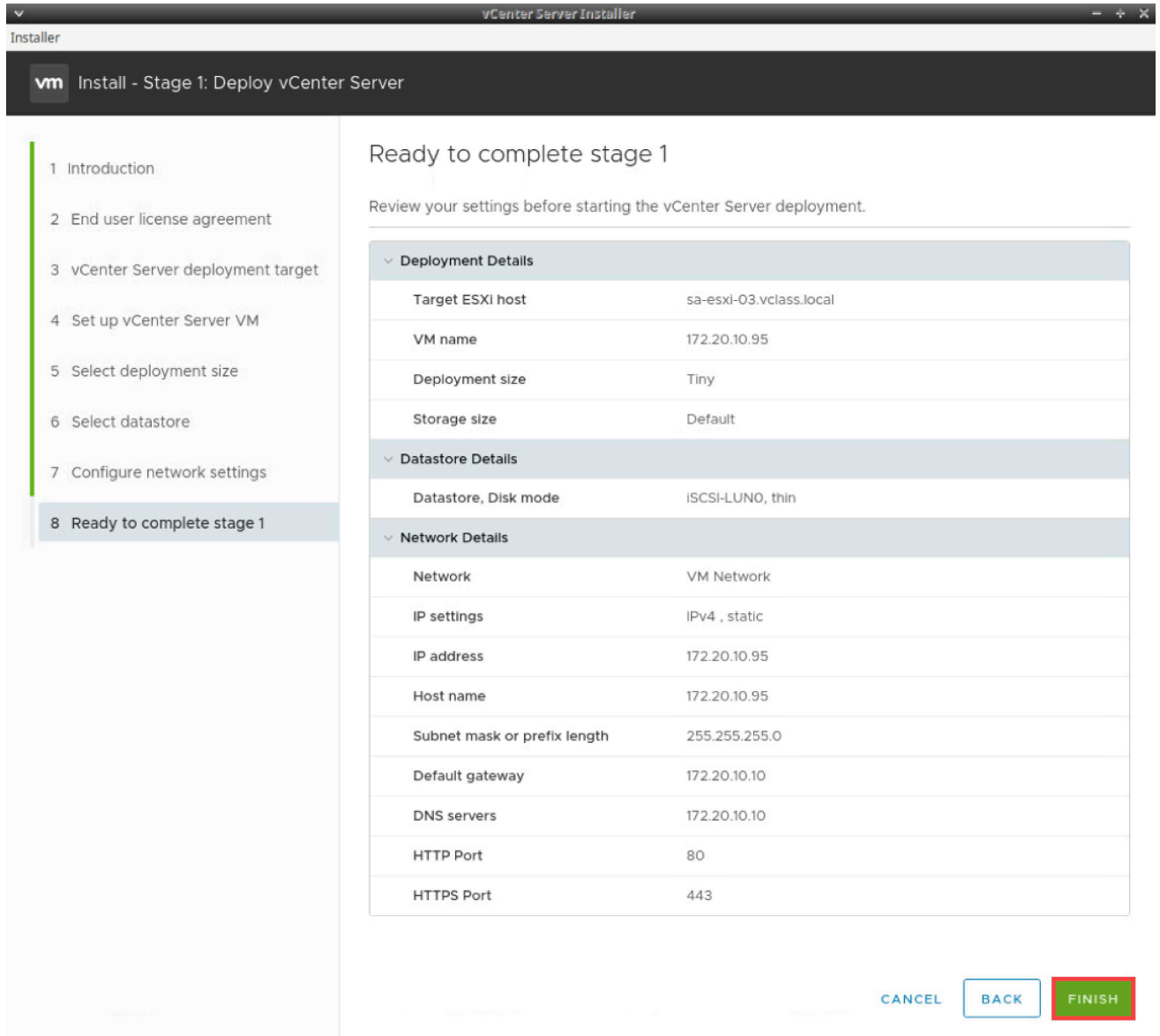
Network	VM Network	ⓘ
IP version	IPv4	⌵
IP assignment	static	⌵
FQDN	172.20.10.95	ⓘ
IP address	172.20.10.95	
Subnet mask or prefix length	255.255.255.0	ⓘ
Default gateway	172.20.10.10	
DNS servers	172.20.10.10	
Common Ports		
HTTP	80	
HTTPS	443	

CANCEL BACK NEXT

16. On the Ready to complete stage 1 page, review the details and click Finish.

Please Note

Due to the deployment process potentially taking 20 minutes, it is advised to continuously monitor the process of stage 1.



Deployment Details	
Target ESXi host	sa-esxi-03.vclass.local
VM name	172.20.10.95
Deployment size	Tiny
Storage size	Default

Datastore Details	
Datastore, Disk mode	ISCSI-LUN0, thin

Network Details	
Network	VM Network
IP settings	IPv4 , static
IP address	172.20.10.95
Host name	172.20.10.95
Subnet mask or prefix length	255.255.255.0
Default gateway	172.20.10.10
DNS servers	172.20.10.10
HTTP Port	80
HTTPS Port	443

CANCEL BACK FINISH

17. Monitor the Install – Stage 1: Deploy vCenter Server window. Once it completes, proceed to the next step.



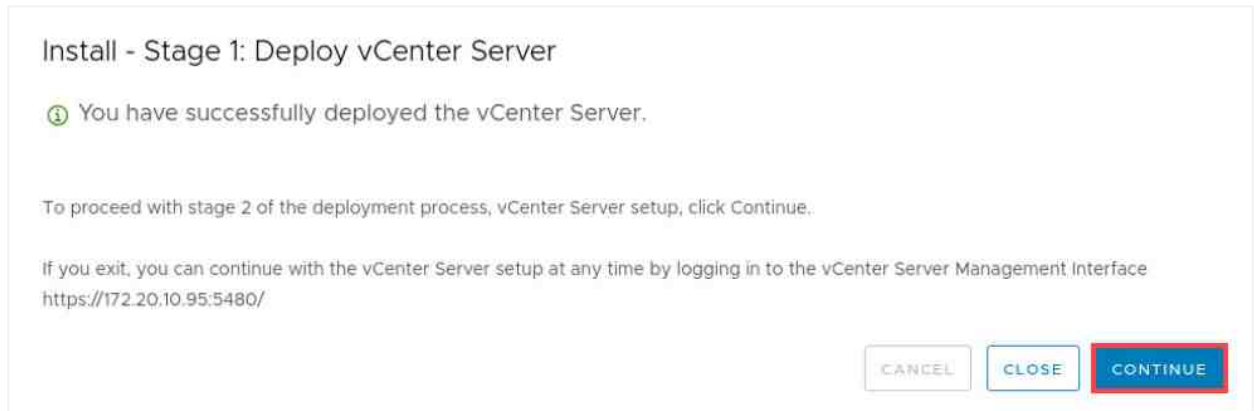
Install - Stage 1: Deploy vCenter Server

0%

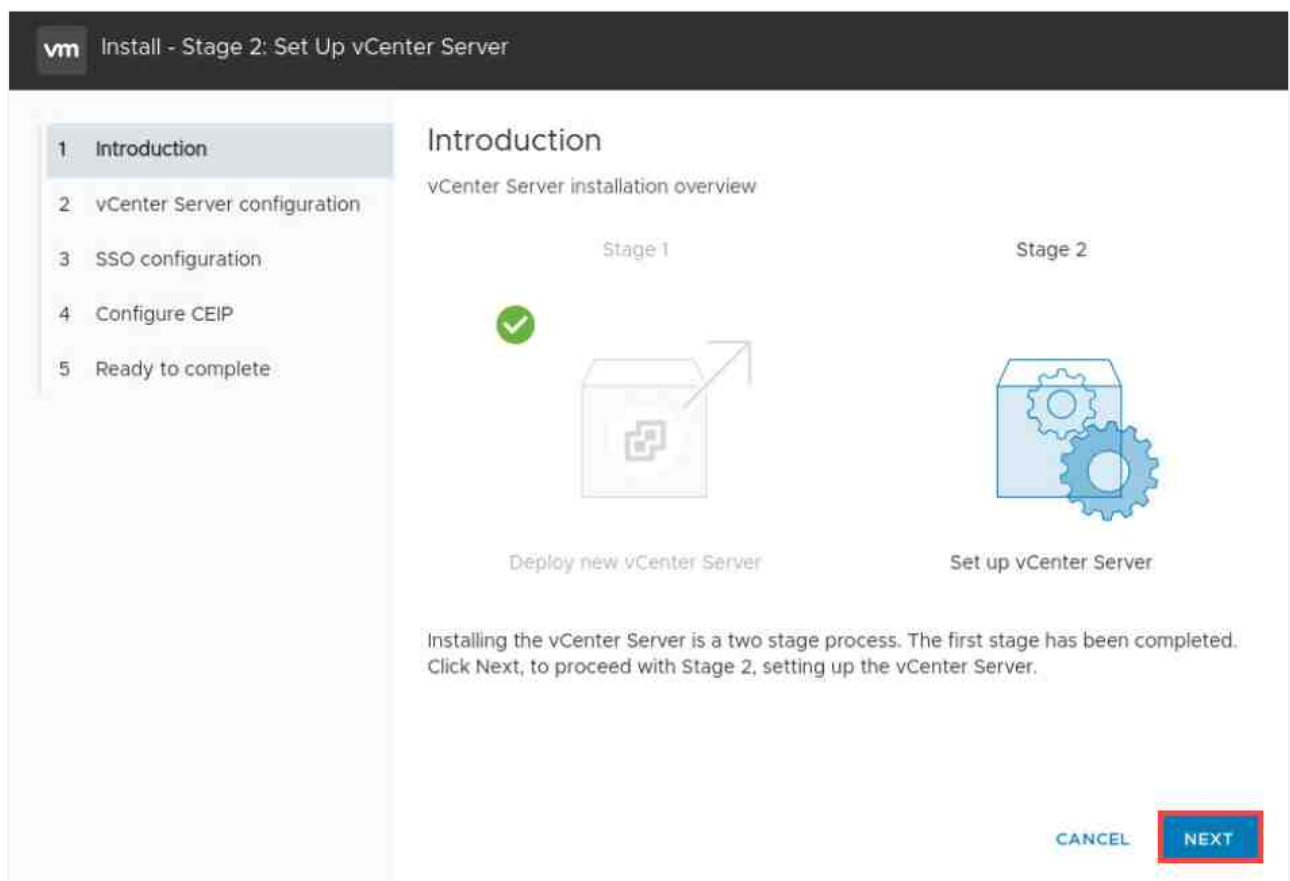
Initializing...

CANCEL

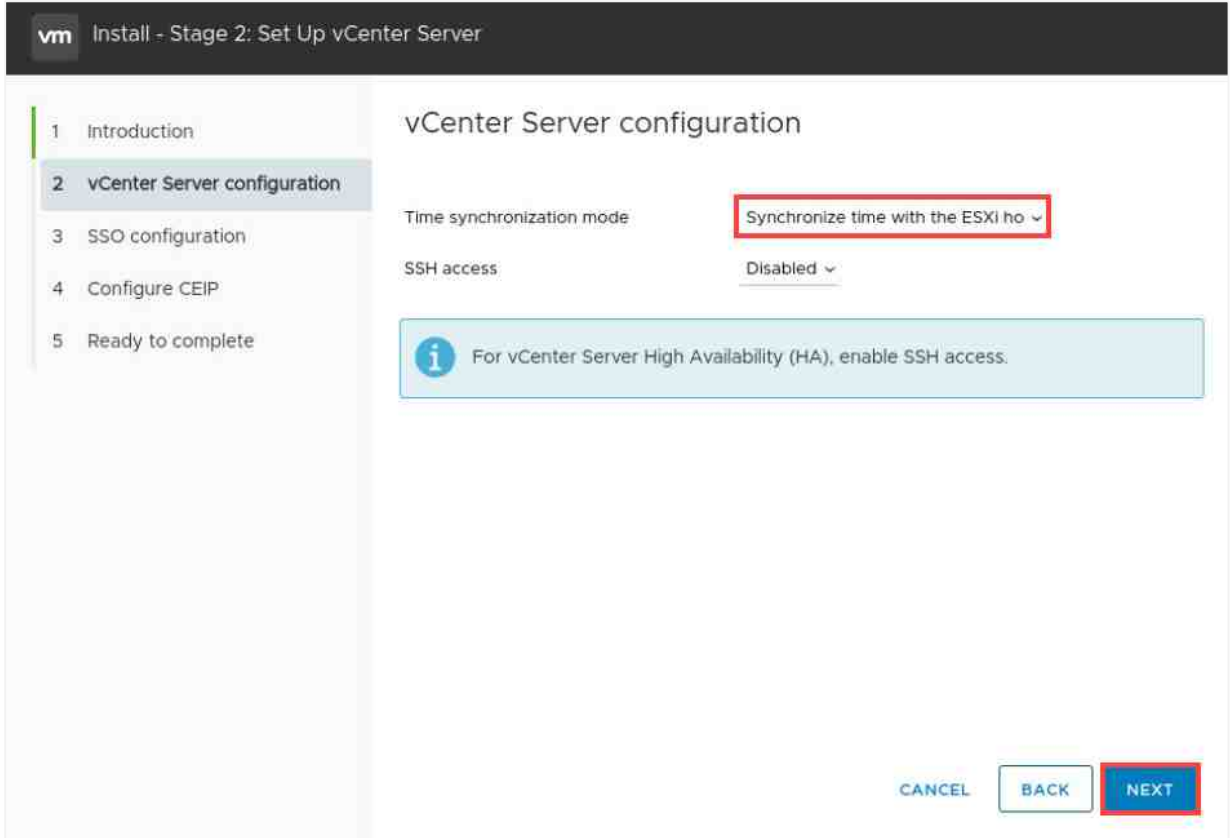
18. Once Stage 1 completes, on the Install – Stage 1: Deploy vCenter Server window, click Continue.



19. Notice the Install - Stage 2 wizard appears. On the Introduction page, click Next to proceed.



20. On the vCenter Server configuration step, verify the Time synchronization mode is set to Synchronize time with the ESXi host. Click Next.



The screenshot shows the 'vCenter Server configuration' step of the VMware vCenter Server installation wizard. The left sidebar contains a list of steps: 1 Introduction, 2 vCenter Server configuration (highlighted), 3 SSO configuration, 4 Configure CEIP, and 5 Ready to complete. The main area is titled 'vCenter Server configuration' and contains two settings: 'Time synchronization mode' set to 'Synchronize time with the ESXi host' (highlighted with a red box) and 'SSH access' set to 'Disabled'. Below these settings is a blue information box with an 'i' icon and the text: 'For vCenter Server High Availability (HA), enable SSH access.' At the bottom right, there are three buttons: 'CANCEL', 'BACK', and 'NEXT' (highlighted with a red box).

vm Install - Stage 2: Set Up vCenter Server

1 Introduction

2 vCenter Server configuration

3 SSO configuration

4 Configure CEIP

5 Ready to complete

vCenter Server configuration

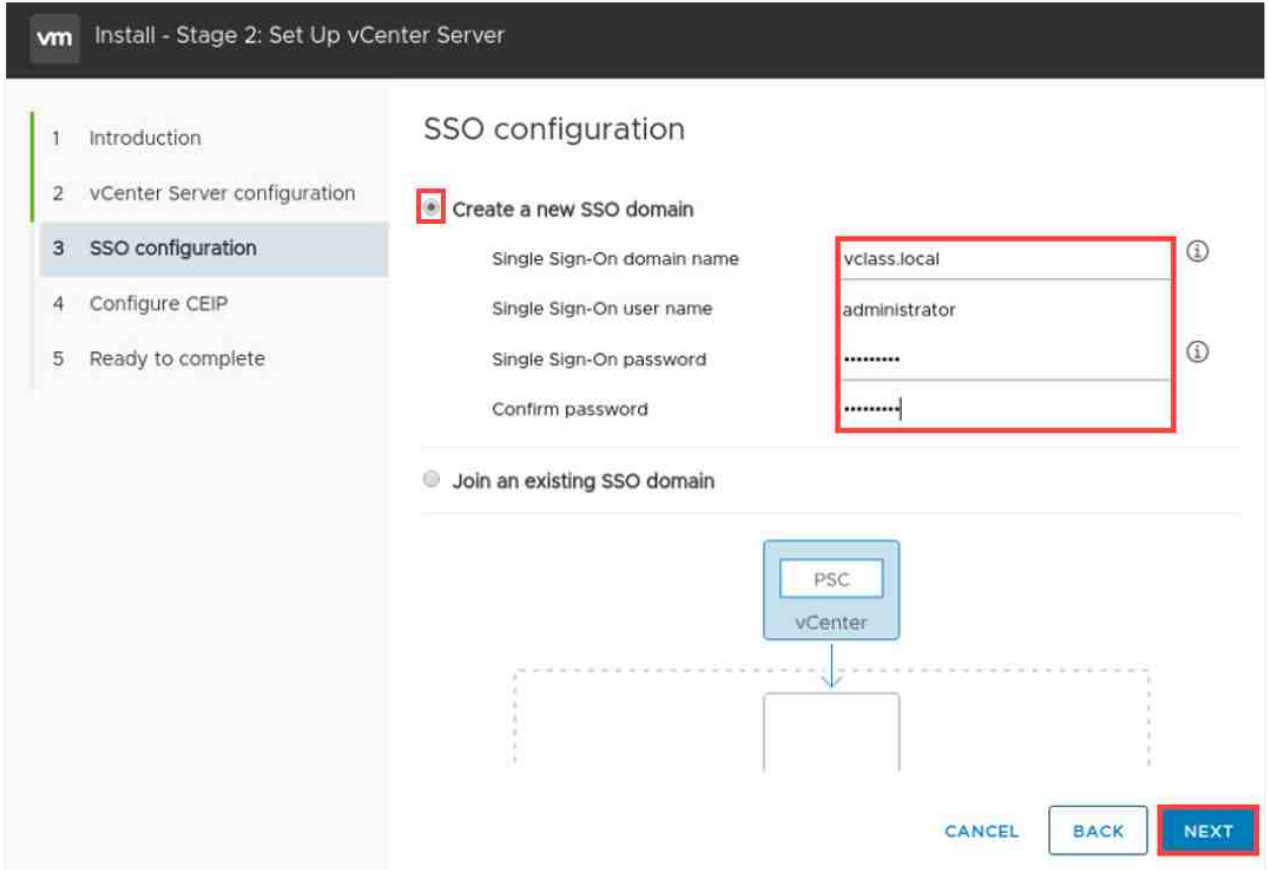
Time synchronization mode: Synchronize time with the ESXi host

SSH access: Disabled

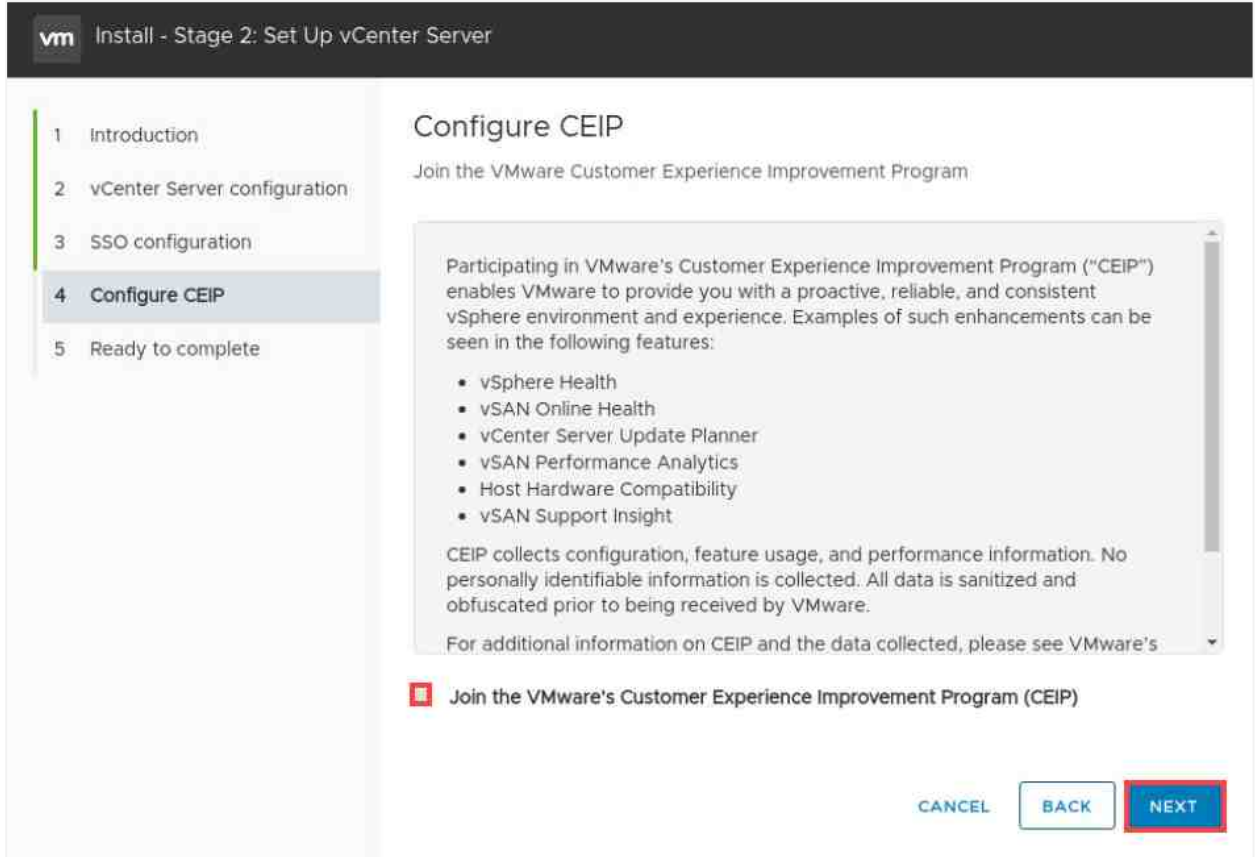
For vCenter Server High Availability (HA), enable SSH access.

CANCEL BACK NEXT

21. On the SSO configuration page, ensure the radio button for Create a new SSO domain is selected and enter the following:
- Single Sign-On domain name: `vclass.local`
 - Single Sign-On password: `Train1ng$`
 - Confirm password: `Train1ng$`
 - Click Next.



22. On the Configure CEIP page, deselect the Join the VMware's Customer Experience Improvement Program. Click Next.



The screenshot shows the VMware vCenter Server installation wizard at Stage 2: Set Up vCenter Server. The left sidebar contains a navigation menu with five items: 1 Introduction, 2 vCenter Server configuration, 3 SSO configuration, 4 Configure CEIP (highlighted), and 5 Ready to complete. The main content area is titled 'Configure CEIP' and includes the sub-header 'Join the VMware Customer Experience Improvement Program'. A scrollable box contains text explaining that participating in VMware's Customer Experience Improvement Program (CEIP) enables VMware to provide a proactive, reliable, and consistent vSphere environment and experience. It lists several features: vSphere Health, vSAN Online Health, vCenter Server Update Planner, vSAN Performance Analytics, Host Hardware Compatibility, and vSAN Support Insight. Below this, it states that CEIP collects configuration, feature usage, and performance information, and that no personally identifiable information is collected. At the bottom of the scrollable box, it says 'For additional information on CEIP and the data collected, please see VMware's'. Below the scrollable box, there is a checkbox labeled 'Join the VMware's Customer Experience Improvement Program (CEIP)'. At the bottom right of the page, there are three buttons: 'CANCEL', 'BACK', and 'NEXT' (highlighted with a red border).

vm Install - Stage 2: Set Up vCenter Server

1 Introduction
2 vCenter Server configuration
3 SSO configuration
4 **Configure CEIP**
5 Ready to complete

Configure CEIP

Join the VMware Customer Experience Improvement Program

Participating in VMware's Customer Experience Improvement Program ("CEIP") enables VMware to provide you with a proactive, reliable, and consistent vSphere environment and experience. Examples of such enhancements can be seen in the following features:

- vSphere Health
- vSAN Online Health
- vCenter Server Update Planner
- vSAN Performance Analytics
- Host Hardware Compatibility
- vSAN Support Insight

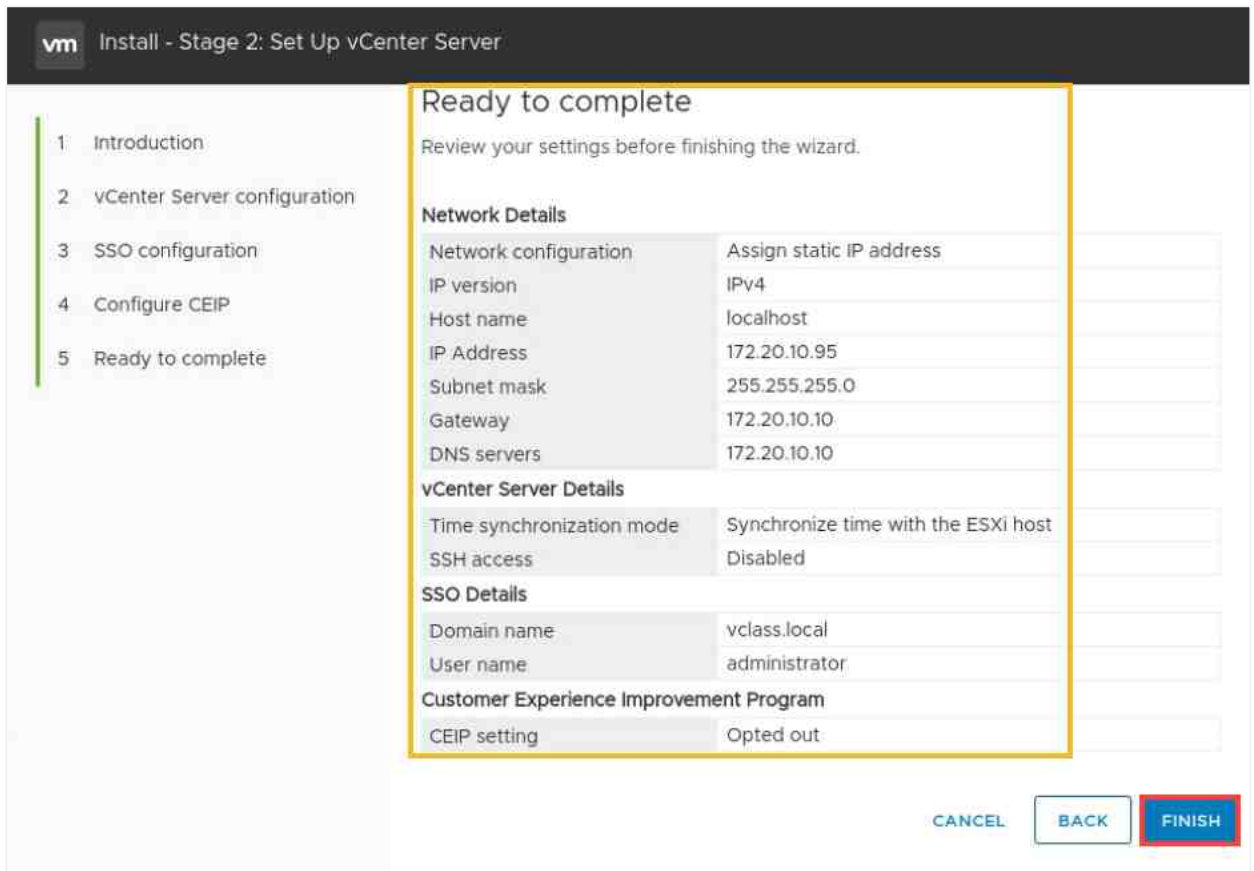
CEIP collects configuration, feature usage, and performance information. No personally identifiable information is collected. All data is sanitized and obfuscated prior to being received by VMware.

For additional information on CEIP and the data collected, please see VMware's

☐ Join the VMware's Customer Experience Improvement Program (CEIP)

CANCEL BACK **NEXT**

23. On the Ready to complete page, review the configurations and click Finish.



The screenshot shows the 'Ready to complete' page of the vCenter Server installation wizard. On the left, a progress bar indicates the current stage is '5 Ready to complete'. The main content area is titled 'Ready to complete' and includes a sub-header 'Review your settings before finishing the wizard.' Below this, there are three sections of configuration details:

Network Details	
Network configuration	Assign static IP address
IP version	IPv4
Host name	localhost
IP Address	172.20.10.95
Subnet mask	255.255.255.0
Gateway	172.20.10.10
DNS servers	172.20.10.10

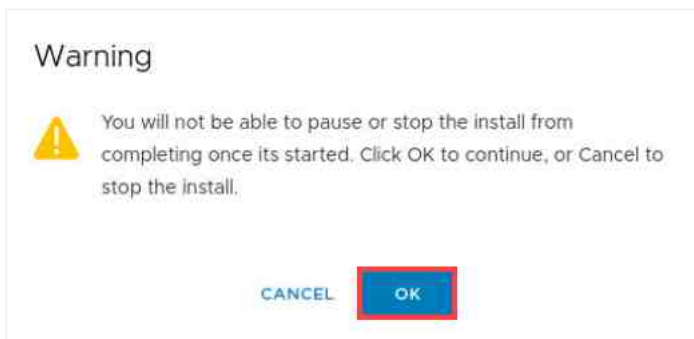
vCenter Server Details	
Time synchronization mode	Synchronize time with the ESXi host
SSH access	Disabled

SSO Details	
Domain name	vclass.local
User name	administrator

Customer Experience Improvement Program	
CEIP setting	Opted out

At the bottom right, there are three buttons: 'CANCEL', 'BACK', and 'FINISH'. The 'FINISH' button is highlighted with a red border.

24. On the Warning window, click OK.

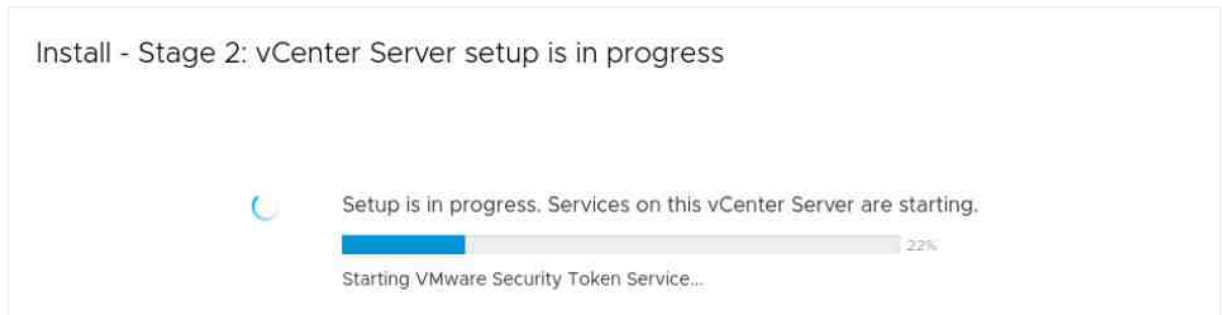


The screenshot shows a 'Warning' dialog box. It contains a yellow warning icon and the following text: 'You will not be able to pause or stop the install from completing once its started. Click OK to continue, or Cancel to stop the install.' At the bottom, there are two buttons: 'CANCEL' and 'OK'. The 'OK' button is highlighted with a red border.

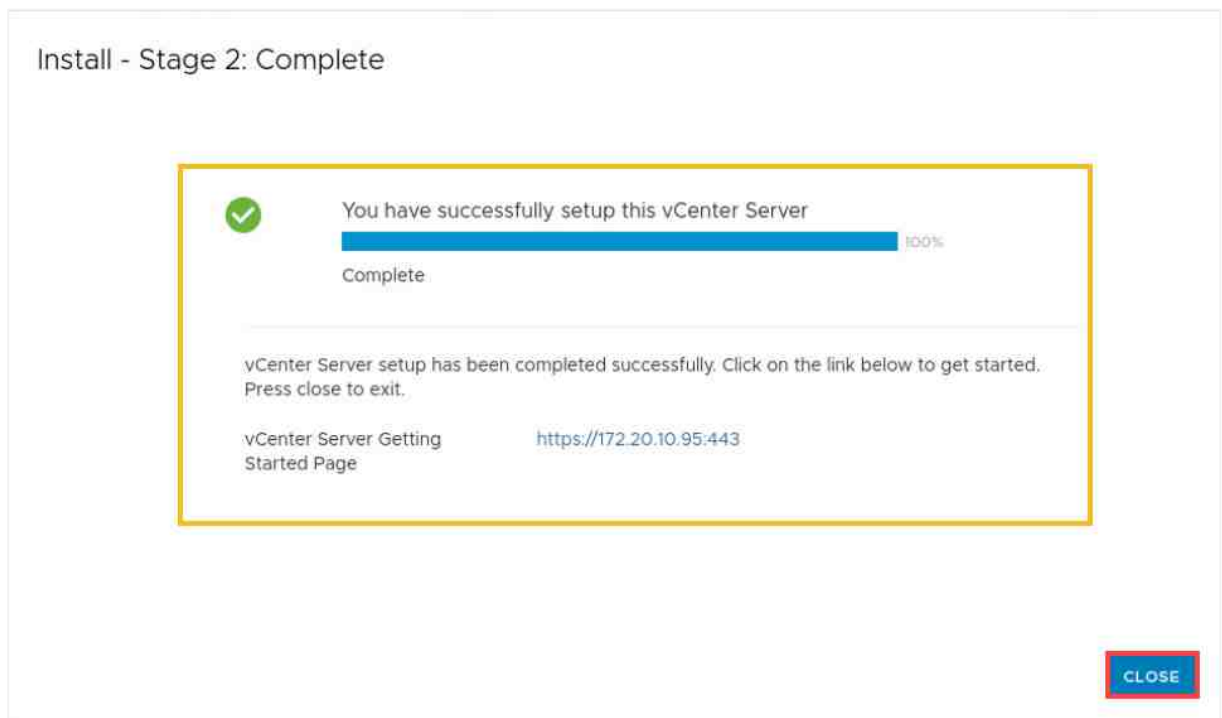
25. Monitor the Install – Stage 2: vCenter Server Setup is in progress window.

**Please
Note**

Stage 2 of the vCenter Server setup will take up to 20 minutes to complete. Monitor the progress by viewing the status window.



26. Once Stage 2 is complete, notice the IP address of your newly created vCenter Server Appliance. Click Close.



27. If you get the Failed to execute default Web Browser error, click Close.



28. Continue to the next task.

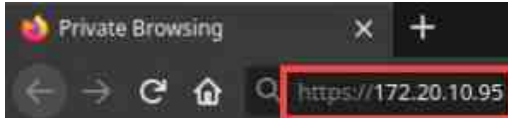
3 Access vCenter Server Appliance

In this task, you will verify access to your new vCenter Server Appliance.

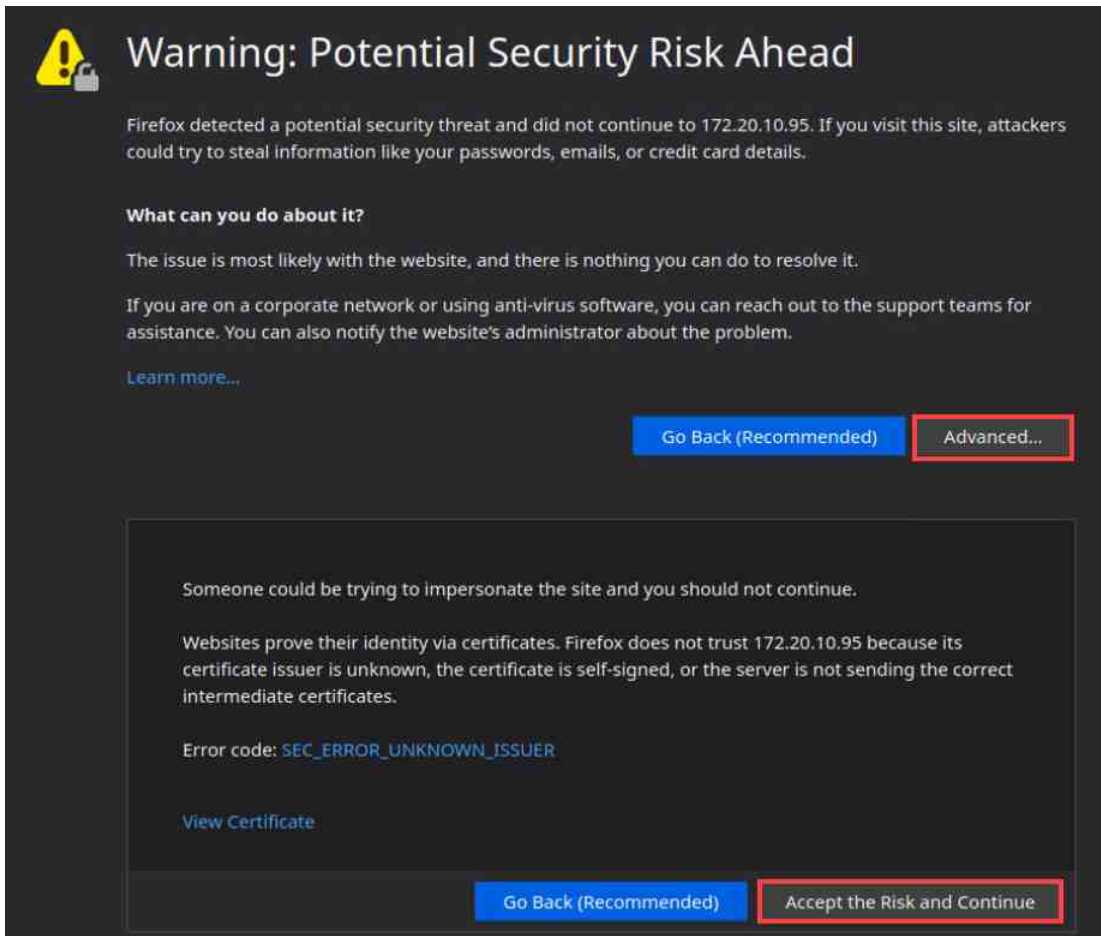
1. Open a new Firefox window by clicking on the Firefox icon located in the taskbar.



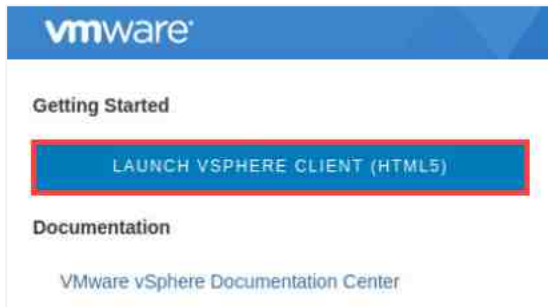
2. In the address field, type `https://172.20.10.95` and press Enter.



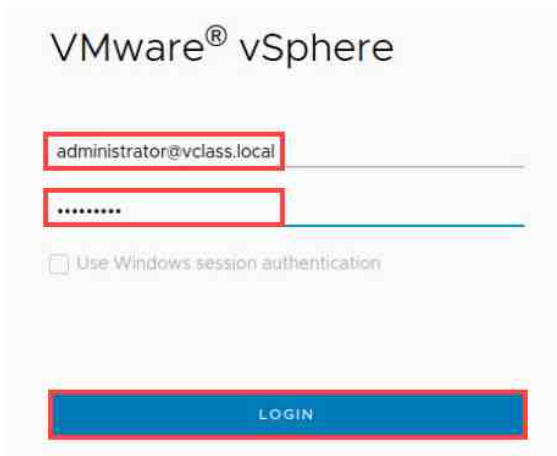
3. On the Warning: Potential Security Risk Ahead window, click Advanced and Accept the Risk and Continue.



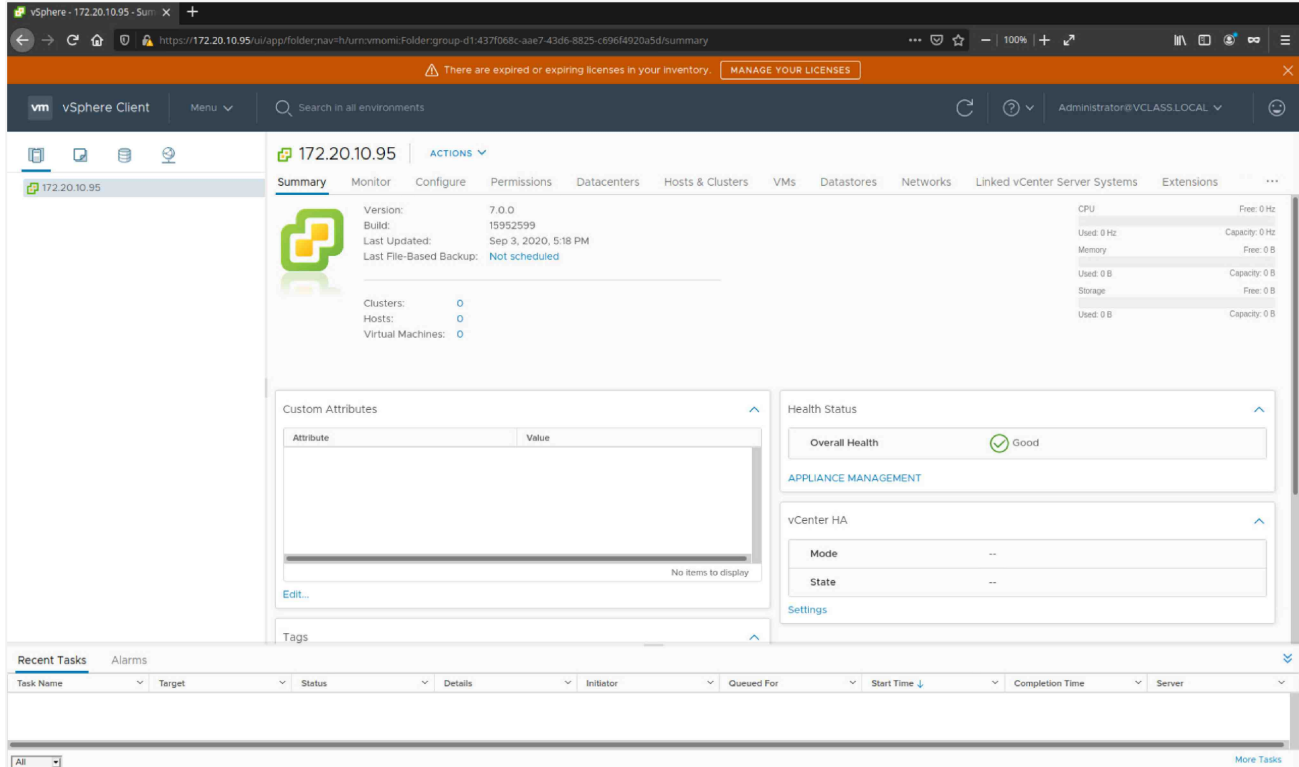
4. Once the VMware Getting Started page appears, click on the Launch vSphere Client (HTML5) button.



5. Log in to vCenter Server using `administrator@vclass.local` as the username and `Training$` as the password.



- Once you have logged in to your newly create vCenter Server, you may navigate vCenter at your convenience.

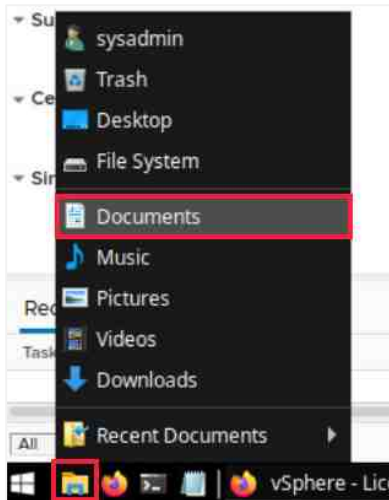


- Leave the vSphere Web Client open and continue to the next task.

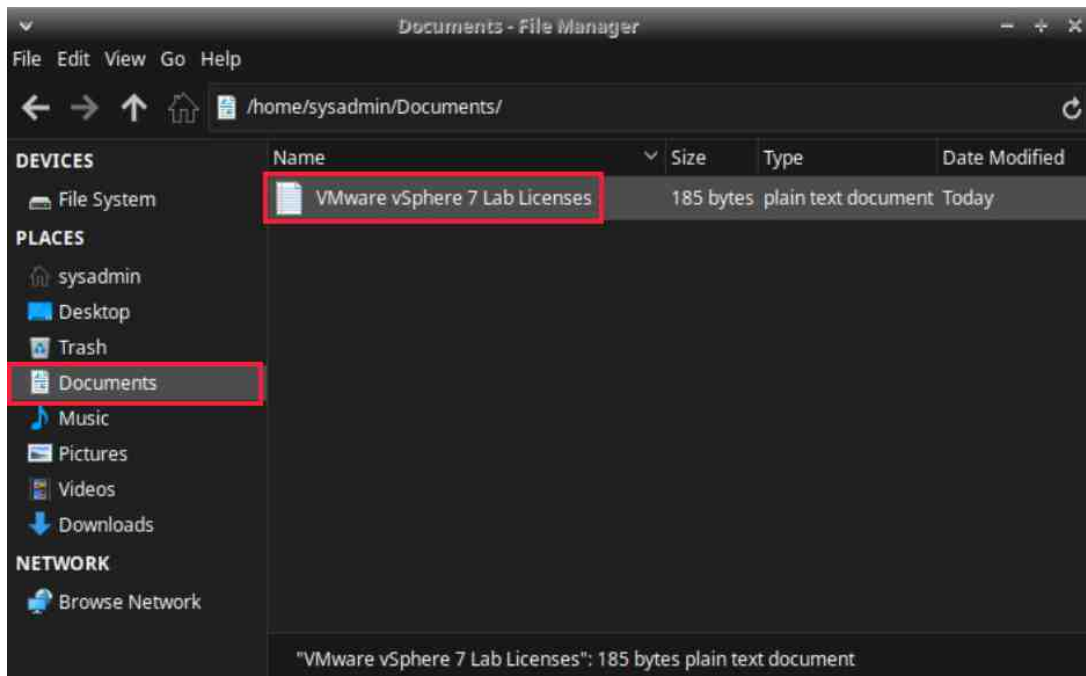
4 License vCenter Server Appliance

In this task, you will add VMware vCenter Server 7 license to VMware vCenter Server.

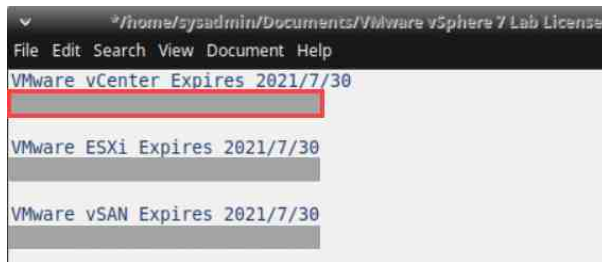
1. Change focus to the sa-student desktop. Open the Places folder and click Documents.



2. In the Documents - File Manager window, open the VMware vSphere 7 Lab Licenses text file.



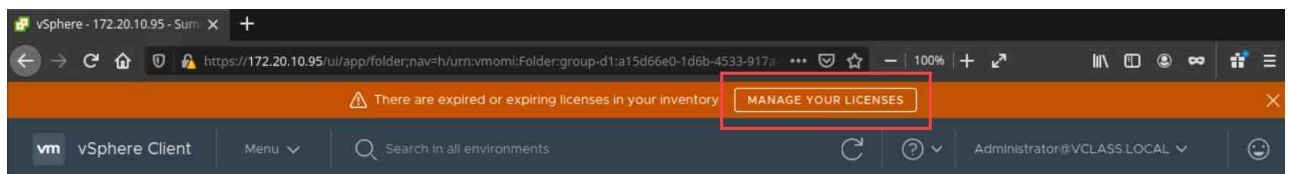
3. In the VMware vSphere 7 Lab Licenses text file, copy the VMware vCenter license key by pressing Ctrl+C.



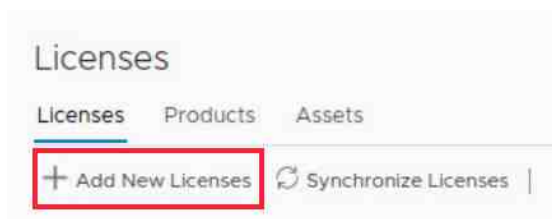
4. Close the VMware vSphere 7 Lab Licenses text file and Documents folder by clicking the x in the upper-right corner of the windows.



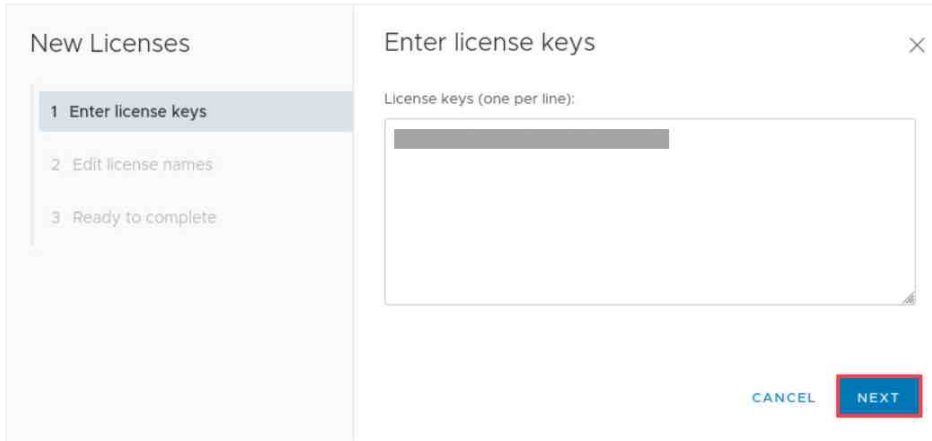
5. On the main vSphere Web Client page, click Manage Your License.



6. Click Add New License.

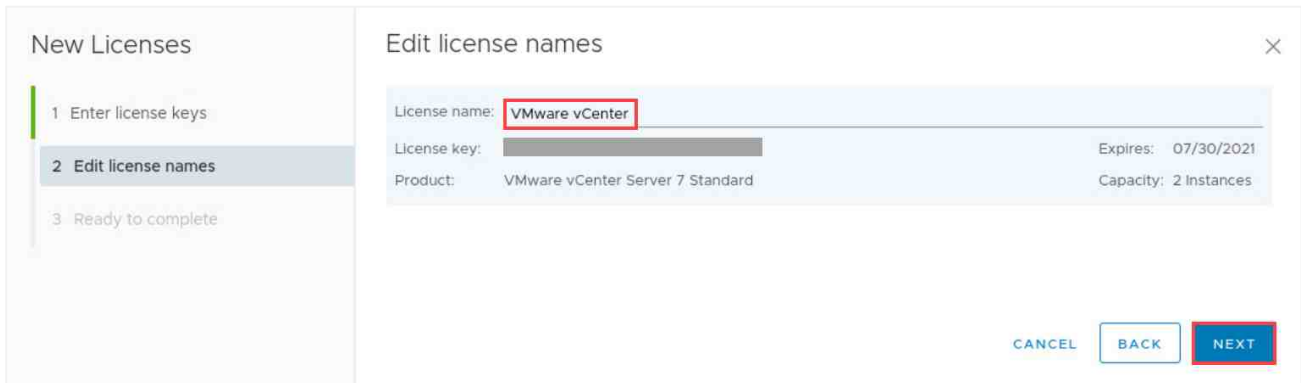


7. In the New Licenses window, on the Enter license keys step, paste (CTRL+V) in the text field and make sure the correct license key has been populated for vCenter Server. Click Next.



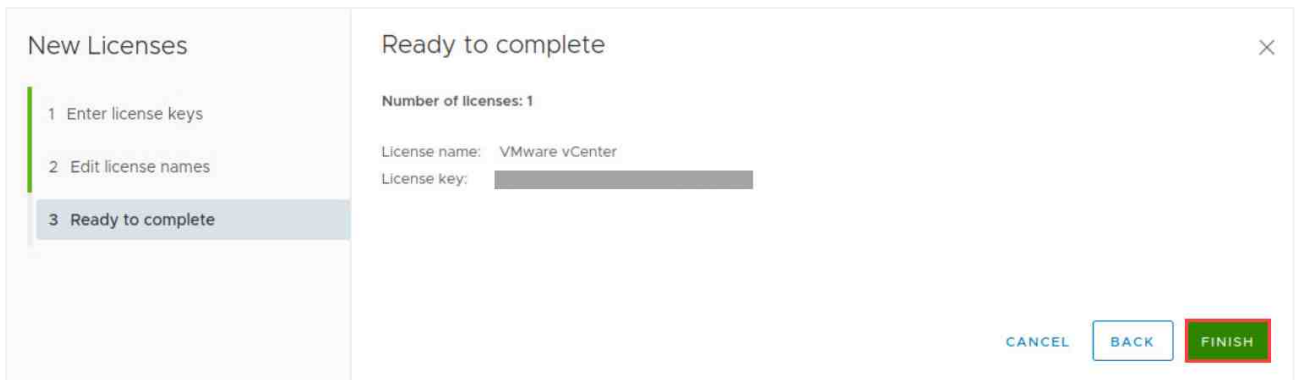
The screenshot shows the 'New Licenses' window with the 'Enter license keys' step selected. The left sidebar shows three steps: '1 Enter license keys' (active), '2 Edit license names', and '3 Ready to complete'. The main area is titled 'Enter license keys' and contains a text field labeled 'License keys (one per line):'. The 'NEXT' button is highlighted with a red box.

8. On the Edit license names step, enter VMware vCenter as the license name for its respective key and click Next.



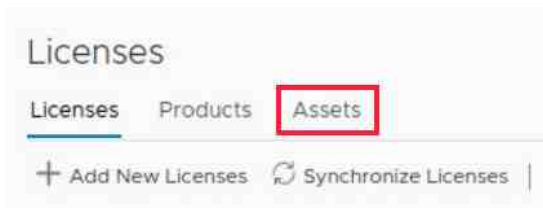
The screenshot shows the 'New Licenses' window with the 'Edit license names' step selected. The left sidebar shows three steps: '1 Enter license keys', '2 Edit license names' (active), and '3 Ready to complete'. The main area is titled 'Edit license names' and contains a form with the following fields: 'License name:' (with 'VMware vCenter' entered and highlighted by a red box), 'License key:', 'Product:' (with 'VMware vCenter Server 7 Standard' entered), 'Expires:' (with '07/30/2021' entered), and 'Capacity:' (with '2 Instances' entered). The 'NEXT' button is highlighted with a red box.

9. On the Ready to complete step, click Finish.

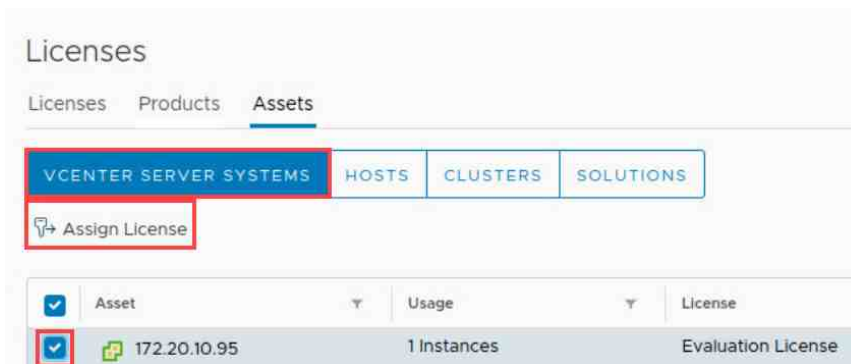


The screenshot shows the 'New Licenses' window with the 'Ready to complete' step selected. The left sidebar shows three steps: '1 Enter license keys', '2 Edit license names', and '3 Ready to complete' (active). The main area is titled 'Ready to complete' and contains the following information: 'Number of licenses: 1', 'License name:' (with 'VMware vCenter' entered), and 'License key:'. The 'FINISH' button is highlighted with a red box.

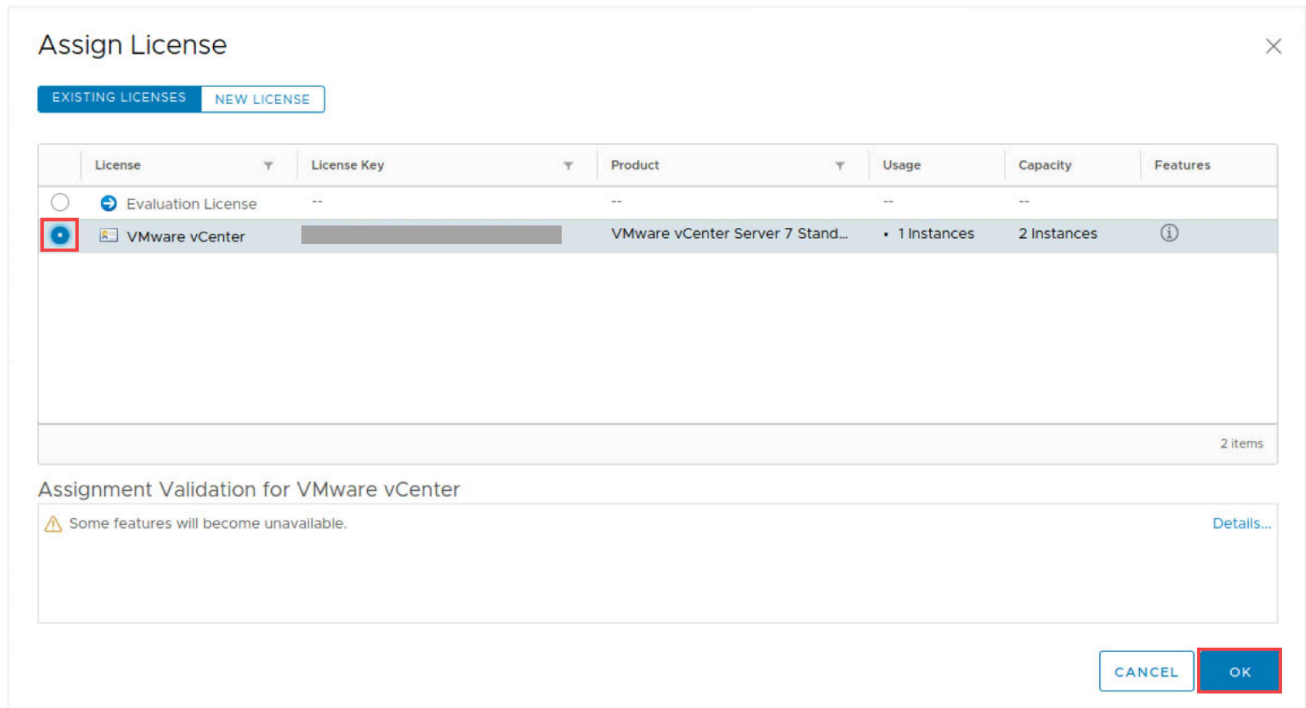
10. In the main workspace, click on the Assets tab.



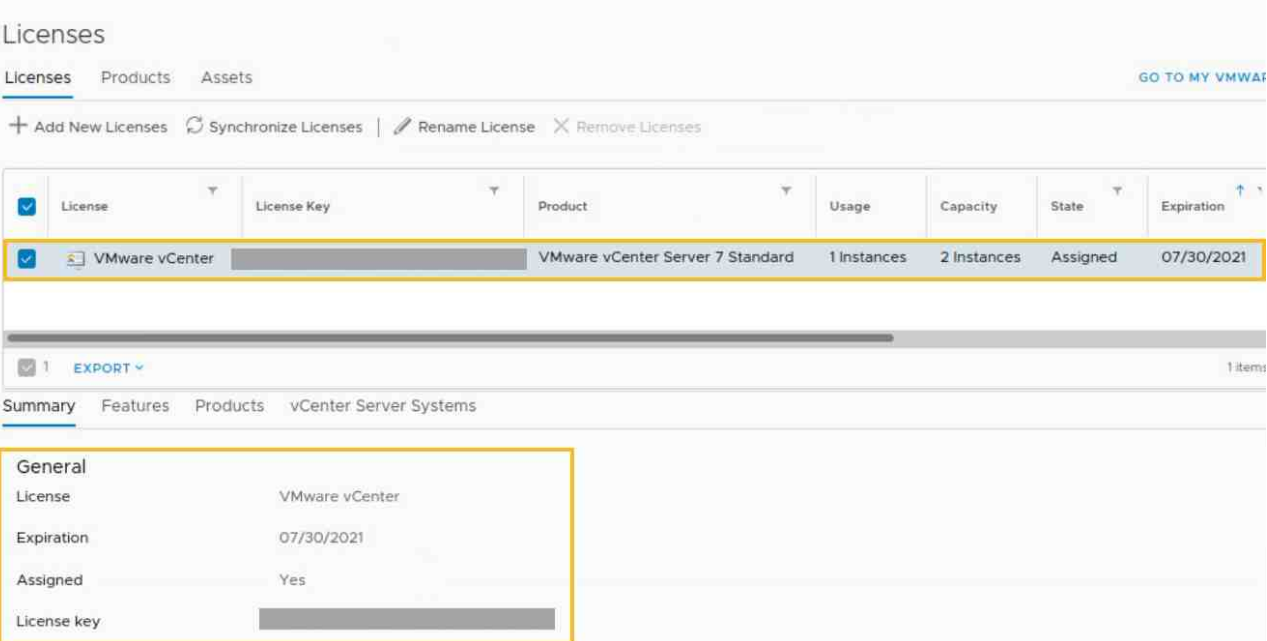
11. On the Assets tab, select the vCenter Server Systems tab and check the checkbox for 172.20.10.95. Click Assign License.



12. In the Assign License window, select the VMware vCenter radio button and click OK.



13. In the main workspace, click on the Licenses tab. Your newly created VMware vCenter Server Appliance is now licensed.



The screenshot shows the VMware vCenter 'Licenses' page. At the top, there are tabs for 'Licenses', 'Products', and 'Assets'. Below the tabs are action buttons: '+ Add New Licenses', 'Synchronize Licenses', 'Rename License', and 'Remove Licenses'. A table lists the licenses with columns: License, License Key, Product, Usage, Capacity, State, and Expiration. One license is listed: 'VMware vCenter' with a license key, 'VMware vCenter Server 7 Standard', '1 Instances', '2 Instances', 'Assigned', and '07/30/2021'. Below the table is an 'EXPORT' button. Underneath the table, there are tabs for 'Summary', 'Features', 'Products', and 'vCenter Server Systems'. The 'Summary' tab is selected, showing a 'General' section with fields: License (VMware vCenter), Expiration (07/30/2021), Assigned (Yes), and License key (redacted).

License	License Key	Product	Usage	Capacity	State	Expiration
VMware vCenter		VMware vCenter Server 7 Standard	1 Instances	2 Instances	Assigned	07/30/2021

EXPORT

Summary Features Products vCenter Server Systems

General

License: VMware vCenter

Expiration: 07/30/2021

Assigned: Yes

License key: [Redacted]

14. The lab is now complete; you may end the reservation.